SUMMARY REFERENCE

Capability Matters

Case Overview

A quick-reference guide to one hundred real cases of capability — how it fails, how it has been engineered, and where the discipline is being tested next — each with a brief callout, its key sources, and the LENS courses it informs.

LDT · LENS — JOHNS HOPKINS UNIVERSITY SCHOOL OF EDUCATION

Learning Engineering for Next-Generation Systems

LENS is a specialization within the Johns Hopkins University School of Education's **Master of Education in Learning Design and Technology** (LDT). Its premise is that capability — the ability of people and institutions to perform demanding, high-consequence work reliably — can be **engineered**: specified, measured, designed for, and improved, rather than left to tradition, heroics, or chance. The program trains practitioners to treat capability as a system parameter, to build the evidence that shows whether an intervention actually worked, and to design the human and technical halves of a system together rather than in isolation.

LENS rests on five pillars: **Mission Literacy** — fluency in the operational domains where capability is contested; the **JHU Ecosystem** — the research, applied-laboratory, and school assets the program draws on; **Intersectional Expertise** — pairing learning science with engineering and domain practice; **Capability Focus** — capability itself as the unit of design; and **Flywheel Iteration** — the Practice Flywheel, the iterative loop from requirements to evidence that animates the other four. The pillars are designed to be load-bearing together, not individually.

One hundred cases, three movements

The casebook assembles **one hundred** documented cases — from healthcare, defense, education, energy, transportation, and government — and organizes them in three parts.

Part I — The Failure Modes groups cases under six recurring patterns (see overleaf), with an evidence-gap chapter on measurement. The taxonomy is a finding, not a theory: six categories account for almost every well-documented preventable failure in the literature.

Part II — What Works collects the successes, which share one structural feature: a **paired intervention** — a technical artifact joined to a cultural authority; a measurement instrument joined to a body that acts on it. Neither half alone moves the curve.

Part III — The Frontier concerns the cases where the discipline is being asked to specify what good looks like before the catastrophe arrives — particularly human-AI teaming. These cases are deliberately less settled.

Reading an entry

Each case is distilled to a single panel with four parts: a **header** (case number, domain tags, and year); the **title**; an ~100-word **callout** that states what happened and why it matters; one to three **key references**; and a short note on its **LENS applicability** — the courses that use it and the capability lesson it carries. The full three-page treatment (an inline-cited narrative and a Learning Engineering Lens) lives in the main casebook; this guide is the index to it.

DOMAIN TAGS

DEFENSE Defense · AVIATION Aviation · SPACE / AEROSPACE Space / Aerospace · HEALTHCARE Healthcare · ENERGY Energy

EDUCATION AT SCALE Education at Scale · GOVERNMENT Government · INDUSTRIAL Industrial · TECHNOLOGY Technology · AUTONOMOUS SYSTEMS Autonomous Systems

FAILURE-MODE CODES

In the main casebook each case carries one or more single-letter mode codes: **T** training gap · **D** capability designed out · **N** normalization of deviance · **H** human-system interface · **G** governance & trust · **K** knowledge & institutional memory. Most cases carry two or three; the primary mode is listed first.

The six failure modes

T **Training gap**

The operator lacked a capability the situation demanded — and the institution had the means to build it but had let it lapse or never delivered it.

D **Capability designed out**

The capability was removed from, or never built into, the system's design — a defect, omission, or a cost-and-packaging decision made upstream.

N **Normalization of deviance**

A known deviation became routine because it had not yet caused harm — until the margin ran out.

H **Human-system interface**

The interface misled the operator: ambiguous cues, hidden modes, or alarms that did not carry the weight the moment required.

G **Governance & trust**

The institution's incentives, oversight, or disclosure produced the harm — the measurement system or authority structure was the failure.

K **Knowledge & institutional memory**

The institution lost, or never built, the knowledge to do the work — through attrition, silos, or thin transfer between people and units.

The ten LENS courses

Each entry's "LENS applicability" names the courses it informs. Approximate case coverage is shown in parentheses.

LEN 1	Principles of learning engineering	~21
LEN 2	Human-AI teaming	~23
LEN 3	Systems integration	~18
LEN 4	Evidence and measurement	~31
LEN 5	Capability analysis	~36
LEN 6	Applied problem-solving	~13
LEN 7	Bias and governance	~49
LEN 8	Knowledge transfer	~45
LEN 9	Computational methods	~10
LEN 10	Capstone studio	~19

Cases are mapped to more than one course where they apply, so the counts sum to more than one hundred. The distribution is itself diagnostic: the literature is richest in governance, knowledge transfer, and capability analysis, and thinnest in computational instrumentation — a gap the studio sequence is built to close.

CONTENTS

The hundred cases

Listed in reading order; the number is the case's catalog number in the main casebook.

CASE 1 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

In the summer of 2017 two U.S. Navy destroyers of the forward-deployed Seventh Fleet collided with merchant ships nine weeks apart, killing seventeen sailors — seven on the Fitzgerald, ten on the John S. McCain. Their crews' seamanship, navigation, and console skills had eroded under a decade of relentless operational tempo and the self-study "training" that replaced the in-person school the Navy cut in 2003. Investigations by the Navy and the NTSB judged both collisions avoidable and traced them to insufficient training and weak oversight; on the McCain, a confusing touch-screen helm let a watch team believe it had lost steering it never actually lost. The case is this book's canonical training gap: stated readiness and real readiness diverged for years, and the system could no longer see the difference.

THE FULL CASE, IN FIVE BEATS

Background — Forward-deployed Seventh Fleet at peak tempo; in-person SWOS school replaced by self-study CD-ROMs in 2003

What Happened — Fitzgerald struck by ACX Crystal off Japan; McCain collided with Alnic MC near Singapore

The Investigation — Navy and NTSB judged both avoidable, citing training shortfalls and a confusing touch-screen helm design

The Capability Gap — Waivers stacked while readiness dashboards stayed green; stated and actual readiness diverged for a decade

Aftermath & Reform — In-person pipeline rebuilt, Ready-for-Sea Assessment stood up, touch-screen helm slated for conventional replacement

KEY REFERENCES

- T. C. Miller, M. Faturechi & R. Rotella, "Years of Warnings, Then Death and Disaster," ProPublica (2019) — the 2003 SWOS-in-a-Box shift.
- GAO, Navy Readiness, GAO-17-809T (Sept. 2017) — 37% of Japan-based warfare certifications expired by June 2017.
- NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018.

LENS APPLICABILITY

LENS treats this as the worked example of LEN 5 (Human Capability Analysis and Requirements) interacting with LEN 4 (Evidence and Measurement). Students reconstruct the capability requirements for underway watchstanding from operational analysis and design the evidence system that would have surfaced the gap before the collisions. LEN 8 covers the institutional dynamics of deferred training.

CASE 3 AVIATION 2009

Air France Flight 447

Air France Flight 447, an Airbus A330, fell into the South Atlantic on 1 June 2009, killing all 228 aboard — the deadliest accident in the airline's history. At cruise altitude the pitot probes iced over, the airspeed readings failed, and the autopilot handed the jet to a crew that had never trained to fly it by hand in that regime. The pilot flying held the nose up into a full aerodynamic stall and never recognized it; the aircraft descended for nearly four and a half minutes. The BEA traced the loss to the airspeed failure, the crew's inappropriate inputs, and a training system that taught stall prevention at low altitude but never stall recovery at altitude — a gap between the trained envelope and the operational one that reshaped global pilot-training rules.

THE FULL CASE, IN FIVE BEATS

Background — Highly automated A330 crossed equatorial storm band carrying a known but unfitted pitot-probe vulnerability
What Happened — Pitot probes iced, autopilot disconnected, sustained nose-up input stalled the jet for four minutes
The Investigation — BEA cited airspeed loss, inappropriate inputs, and training that never produced reliable stall responses
The Capability Gap — Simulators could not reproduce high-altitude stall; reliable automation had also let manual skills atrophy
Aftermath & Reform — Pitot probes replaced; regulators mandated Upset Prevention and Recovery Training across all airlines

KEY REFERENCES

- Bureau d'Enquêtes et d'Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending.
- BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack.
- BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall.

LENS APPLICABILITY

LENS uses AF447 in LEN 5 to teach the difference between the trained capability envelope and the operational capability envelope, and in LEN 2 to introduce human-AI handoff as a capability problem: the autopilot disconnection was a transition the crew had been trained to avoid rather than to handle. The case maps directly to canonical problem type three — Capability Cliff at Automation Boundary.

CASE 5 ENERGY 1979

Three Mile Island

On 28 March 1979 a small cooling fault at Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, escalated into a partial core meltdown — the most serious accident in U.S. commercial nuclear history. A relief valve stuck open while a control-room light reported it closed, and operators trained for dramatic design-basis ruptures misread the slow, ambiguous cascade and cut the cooling the core needed. A nearly identical near-miss at Davis-Besse eighteen months earlier had never been propagated to the fleet. The Kemeny Commission concluded the fundamental problems were people-related, not equipment. Off-site radiation was minimal, but the accident produced a system of reform — most enduringly INPO — making it the book's paired example of a failure that engineered lasting capability.

THE FULL CASE, IN FIVE BEATS

Background — Operators trained for design-basis ruptures; a near-identical Davis-Besse precursor was never propagated to the fleet

What Happened — Relief valve stuck open while panel reported closed; operators throttled injection the core needed

The Investigation — Kemeny Commission concluded fundamental problems were people-related, reaching past operators to management and the NRC

The Capability Gap — Training mismatched the failure that arrived; capacity to learn from precursors had broken down

Aftermath & Reform — INPO formed within nine months; NRC required plant-referenced simulators and tied training to national standards

KEY REFERENCES

- Kemeny Commission, Report of the President's Commission on the Accident at Three Mile Island (Oct. 1979) — operator training oriented to large design-basis accidents.
- M. Rogovin & G. T. Frampton, Three Mile Island: A Report to the Commissioners and to the Public, NUREG/CR-1250 (U.S. NRC, 1980) — the September 1977 Davis-Besse stuck-PORV precursor and the failure to disseminate its lessons.
- U.S. Nuclear Regulatory Commission, Backgrounder on the Three Mile Island Accident — accident sequence, misleading PORV indication, throttled high-pressure injection, 50% core damage, minimal off-site dose.

LENS APPLICABILITY

LENS uses TMI in LEN 1 as the foundational problem-framing case: when trained scenarios and operational scenarios diverge, the divergence is the real risk. LEN 5 examines the control-room interface and the gap between what indicators displayed and what operators inferred; the pairing with INPO (Case 16) is the Failure-to-Reform arc this book treats as its strongest argument.

CASE 9 DEFENSE ONGOING

Marine Corps Training in the INDOPACOM AOR

The 2022 National Defense Strategy names the Indo-Pacific the Pentagon's priority theater and China its "pacing challenge" — yet the theater so designated has among the least mature live-training infrastructure in the force. For nearly a decade the U.S. Marine Corps has been unable to meet its training requirements at Indo-Pacific ranges, papering over the gap with rotations back to U.S. ranges, virtual substitutes, and multinational exercises pressed into proxy duty. In May 2024 the GAO documented the decade-long shortfall and urged the Corps to analyze its unmet requirements and build a remediation plan. The case is this book's live, ongoing counterpart to its historical failures: a fully recognized, repeatedly documented capability gap that declared priority has not closed, because engineered priority — ranges, basing, certified units — is slow and expensive.

THE FULL CASE, IN FIVE BEATS

Background — Strategy names the Indo-Pacific top priority while its live-training infrastructure remains among the least mature
What Happened — Marine Corps papered over unmet range requirements with rotations, virtual substitutes, and multinational exercises
The Investigation — GAO documented decade-long unmet requirements in 2024; the Department concurred, echoing diagnoses pressed for years
The Capability Gap — Declared priority is cheap; engineered priority follows where construction and dollars actually flow
Aftermath & Reform — GAO recommendations remain open; closure depends on programmed ranges, dollars, and schedule, not another document

KEY REFERENCES

- 2022 National Defense Strategy of the United States of America (U.S. Department of Defense, 2022) — the Indo-Pacific as priority theater and China as the "pacing challenge."
- U.S. Government Accountability Office, Military Readiness: Actions Needed for DOD to Address Challenges, GAO-24-107463 (May 2024) — the least-mature training infrastructure in the priority theater.
- GAO-24-107463 (2024) — the Marine Corps unable to meet INDOPACOM training requirements for nearly a decade, and the CONUS-rotation, virtual, and multinational-exercise workarounds.

LENS APPLICABILITY

LENS uses INDOPACOM in LEN 5 to teach the gap between stated requirements and engineered requirements, and in LEN 8 to examine the organizational dynamics that allow declared priorities to coexist with unfunded capability gaps for a decade. The case is also a live test for any student's claim about how to engineer cross-service capability at scale.

CASE 26 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

The F-35 is the most expensive weapons program in history — roughly 2,500 jets planned and a lifecycle cost above $1.7 trillion, about $1.3 trillion of it in operating and support. The hard part was never the airplane but keeping a global fleet ready, and that half has lagged from the start. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal, with more than 10,000 components in the repair queue and depots behind schedule. GAO traced the shortfall to maintainer shortages, the military's lack of access to technical data, and contractor dependency, and urged a full reassessment of the sustainment strategy. It is the book's cleanest case of a platform fielded faster than the capability infrastructure to sustain it.

THE FULL CASE, IN FIVE BEATS

Background — Most expensive program in history; most of its lifecycle cost is decades of sustainment work
What Happened — Fleet ran at half goal with over 10,000 components queued and depots still behind schedule
The Investigation — GAO traced shortfall to maintainer shortages, lack of access to technical data, and contractor dependency
The Capability Gap — Aircraft fielded faster than maintainers and data; contractor controls knowledge needed to keep jets flying
Aftermath & Reform — GAO urged full sustainment reassessment; costs still rising and fleet readiness still below program goals

KEY REFERENCES

- U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding $1.7 trillion, $1.3 trillion of it in operating and support.
- GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 72-day average depot turnaround; depot stand-up behind schedule.
- GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency.

LENS APPLICABILITY

LENS treats the F-35 in LEN 5 as the canonical case of **Capability- System Misalignment at Transition**, the program's first canonical problem type. Students design the capability infrastructure that should have accompanied each lot of aircraft delivered, including the training pipeline, technical-data deliverables, and depot establishment.

CASE 30 AVIATION 1989

Kegworth / British Midland 92

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400, crashed on the M1 motorway embankment near Kegworth, killing 47 and seriously injuring 74. After a fan blade fractured in the left engine, the crew shut down the right engine — the one still working. Reasoning from older 737s, on which the right engine fed the cabin air, they misattributed the smoke; new, harder-to-read electronic engine displays did not correct them, and the cabin crew who saw flames never told the flight deck. The pilots' mental model was right for the airplane they had flown the week before, but the brief conversion course never overwrote it. The AAIB issued 31 recommendations; Kegworth became the textbook case of capability degrading under system change.

THE FULL CASE, IN FIVE BEATS

Background — New 737-400 variant with redesigned cockpit; conversion was brief and largely self-directed, communicated as reading

What Happened — Fan blade fractured left engine; crew shut down the right reasoning from older 737 model

The Investigation — AAIB cited misapplied mental model, an unreadable vibration indicator, and the silent cabin-to-flight-deck channel

The Capability Gap — Variant treated as incremental change; manual notes never overwrote the prior reflex under emergency pressure

Aftermath & Reform — AAIB issued 31 recommendations spanning conversion training, instrument design, CRM, and crashworthiness

KEY REFERENCES

- Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME.
- AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured.
- AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration.

LENS APPLICABILITY

LENS treats Kegworth in LEN 5 as the canonical **system-change** capability problem and in LEN 8 as the institutional version: how knowledge about what has changed travels from engineering to operations and what gets lost in transit. The case sits alongside Patriot (Case 19) in the canonical problem-type pair for system transition.

CASE 33 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

Friendly fire killed an unusual share of coalition forces in the 1991 Gulf War: 35 of 146 U.S. combat deaths (24%) and 72 of 467 wounded (15%). (The often-cited "2% historical baseline" from Shrader's 1982 study is contested — later estimates run nearer 10–15%, and Shrader stepped back from it.) Post-war reviews blamed the chaos of combat, weak situational awareness and fire-control discipline, and combat-identification failures — and noted the military lacked a shared record to even study its own pattern. Fratricide is the failure of several systems to integrate; despite a generation of combat-ID investment, recurrences in Afghanistan and after show the rate never fell to a confidently low level, because integration is the hardest property to engineer by program.

THE FULL CASE, IN FIVE BEATS

Background — Shrader's contested under-2% baseline became the yardstick that made any later rate look catastrophic

What Happened — A quarter of U.S. KIA died to friendly fire; A-10s struck Marines and British Warriors

The Investigation — Reviews cited combat chaos, weak awareness, fire-control lapses, and the absence of a shared incident database

The Capability Gap — Fratricide is integration across many systems; no single office owns what keeps the rate down

Aftermath & Reform — Better IFF and blue-force tracking followed, yet rates never settled at a confidently low level

KEY REFERENCES

- C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%).
- "Friendly Fire: Facts, Myths and Misperceptions," U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% "historical norm."
- Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, "They Didn't Have to Die". (Per-incident casualty figures vary across sources; see AUDIT.)

LENS APPLICABILITY

LENS uses fratricide in LEN 5 to teach systems-of-systems capability analysis and in LEN 2 to introduce combat identification as a Human-AI Teaming problem (operators rely on automated IFF systems whose limitations are not in their training). LEN 8 examines why decades of awareness have not produced a sustained reduction.

CASE 34 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

After the 1984 death of Libby Zion focused decades of concern on resident-physician fatigue, the ACGME capped resident work at 80 hours a week in 2003 and limited first-year shifts to 16 hours in 2011. The logic was intuitive: tired doctors err, so cut the hours. But hours were one input to a many-variable system; cutting them multiplied error-prone hand-offs and cost residents continuity and procedural repetitions. Two large randomized trials — FIRST (2016) and iCOMPARE (2019) — found flexible schedules non-inferior to the strict caps on patient outcomes, so the promised safety gain never appeared, and in 2017 the ACGME relaxed the intern cap. The case is the book's clearest example of a single-variable intervention into a capability system — and of evidence catching up with a well-meant policy.

THE FULL CASE, IN FIVE BEATS

Background — Libby Zion's 1984 death made resident fatigue a decades-long argument; Bell Commission set early limits
The Intervention — ACGME capped resident work at 80 hours weekly and limited first-year shifts to 16 hours
How It Worked — Cutting hours multiplied hand-offs and cost continuity; trainees often felt less prepared, not rested
The Evidence — FIRST and iCOMPARE found flexible schedules non-inferior; ACGME relaxed the intern cap in 2017
What Transferred — Keystone, CRM, and the surgical checklist engineered supervision and hand-offs alongside the behavioral change

KEY REFERENCES

- B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform.
- Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit.
- D. A. Asch et al., "Resident Duty Hours and Medical Education Policy," NEJM 370: 1671–1673 (2014); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs.

LENS APPLICABILITY

LENS uses duty-hour reform in LEN 5 as the foundational capability-system case (what was the long-hour regime **producing** that was lost when the input was capped?), in LEN 4 to discuss measurement architecture (what FIRST and iCOMPARE measured, and what they did not), and in LEN 10 as a studio prompt for the integrated resident-training redesign that the reforms did not deliver.

CASE 43 AVIATION 2009

Colgan Air Flight 3407

On approach to Buffalo in February 2009, the captain of Colgan Air 3407 responded to a stall warning by pulling back on the controls instead of lowering the nose; the Bombardier Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The NTSB found the captain had a documented history of training failures, and the first officer was ill, fatigued, and paid about $16,000 a year. The airline and the hiring pipeline knew the training history; the regulator did not. Victims’ families mounted one of the most effective aviation-safety campaigns in a generation, producing the 2010 law that raised first-officer experience requirements to 1,500 hours and created the Pilot Records Database. The capability gap was visible to everyone except the system that licensed the pilots.

THE FULL CASE, IN FIVE BEATS

Background — Captain's documented training failures known to the carrier but not to the regulator
What Happened — On approach to Buffalo the captain pulled back at the stick shaker and stalled
The Investigation — NTSB cited fatigue, weak training, and a hiring system blind to the captain's history
The Capability Gap — Pilot training history existed in files but never reached the hiring or licensing decision
Aftermath & Reform — Families drove the 2010 law raising hours to 1,500 and creating the Pilot Records Database

KEY REFERENCES

- NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted).
- NTSB/AAR-10/01 (2010) — the captain’s training history, and the first officer’s fatigue and pay.
- NTSB/AAR-10/01 (2010) — the information-flow gap between known pilot history and the hiring decision.

LENS APPLICABILITY

LENS uses Colgan in LEN 4 as a case for evidence-system design (the PRD as a designed information flow) and in LEN 8 as a case for advocacy-driven institutional change. Studio projects examine how families of victims became the load-bearing element of a reform the industry had resisted for years.

CASE 44 AVIATION 2013

Asiana Airlines Flight 214

On a clear afternoon at San Francisco in July 2013, the crew of Asiana 214 allowed their Boeing 777 to slow far below approach speed without recognizing that the autothrottle was not maintaining it. The aircraft struck the seawall short of runway 28L, killing three of the 307 aboard and injuring nearly two hundred. The NTSB found the crew had mismanaged the approach and inadequately monitored airspeed — but also that the complexity of Boeing's autoflight systems contributed: after the pilots disconnected the autopilot and pulled thrust to idle, the autothrottle reverted to a HOLD mode in which it would not wake to hold the selected speed. The captain believed it would. Asiana 214 is the most prominent recent case of automation surprise on a Western wide-body airliner.

THE FULL CASE, IN FIVE BEATS

Background — Captain new to the 777 hand flew a visual approach with the glideslope out of service

What Happened — Autothrottle sat in HOLD; airspeed decayed silently to 103 knots and the 777 struck the seawall

The Investigation — NTSB cited mismanaged approach, weak airspeed monitoring, and a faulty mental model of the automation

The Capability Gap — The mode the autothrottle was in and the mode the crew believed diverged silently without annunciation

Aftermath & Reform — Recommendations targeted autoflight training, energy state monitoring, and mode annunciation design

KEY REFERENCES

- NTSB, Aircraft Accident Report: Asiana Airlines Flight 214, NTSB/AAR-14/01 (2014) — probable cause and contributing factors (quoted).
- NTSB/AAR-14/01 (2014) — the autothrottle HOLD reversion and airspeed decay to 103 kt.
- NTSB/AAR-14/01 (2014) — the captain's mental model of the autothrottle.

LENS APPLICABILITY

LENS treats Asiana 214 in LEN 2 as a worked case for automation transparency requirements. Students design the interface and training intervention that would have made the mode disconnect salient to the crew at the moment the aircraft slowed.

CASE 45 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

The U.S. Navy entered World War II with the Mark 14 torpedo, so expensive that the Bureau of Ordnance had effectively forbidden live testing in peacetime. Through 1942 submarine crews reported torpedoes running deep, failing to detonate, or exploding prematurely; the Bureau insisted the weapon was sound and blamed the operators. It took eighteen months and the personal intervention of Admiral Charles Lockwood — who ordered fleet-level live-fire tests — to confirm three separate defects: the torpedo ran about ten feet too deep, its magnetic exploder fired erratically, and its contact pin crushed on a square hit. The fixes were simple once the defects were acknowledged. The binding constraint was institutional: a bureau insulated from the operators who knew the weapon did not work.

THE FULL CASE, IN FIVE BEATS

Background — The Bureau of Ordnance had effectively forbidden live trials, so the Mark 14 went to war unproven

What Happened — Submarine crews reported deep runs, premature detonations, and crushed contact pins from early 1942

The Investigation — Lockwood ordered fleet tests; Tinosa's eleven duds on a stopped target finally separated the three defects

The Capability Gap — The binding gap was a channel by which the bureau could be made to hear what the boats already knew

Aftermath & Reform — By late 1943 the three defects were corrected; the episode became the canonical insulated bureau case

KEY REFERENCES

- Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan — operator reports and the Bureau's resistance (paraphrased).
- Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II — testing policy and the three defects.
- Blair (1975) — the USS Tinosa's July 1943 attack (eleven consecutive duds on a stopped target) and Lockwood's fleet-level testing.

LENS APPLICABILITY

LENS uses the Mark 14 in LEN 7 as a governance failure case and in LEN 8 as an organizational-learning case in which the operators had the diagnosis and the institution lacked the structure to receive it. Studio projects (LEN 10) examine what the equivalent operator-to-institution feedback channel should look like.

CASE 46 DEFENSE 1980

Operation Eagle Claw

The April 1980 mission to rescue fifty-two American hostages held at the U.S. embassy in Tehran was aborted at a desert staging point when three of eight helicopters were lost to mechanical and weather problems. During the withdrawal a helicopter collided with a refueling C-130, killing eight servicemembers; the hostages remained captive for another nine months. The Holloway Commission found the operation had been assembled ad hoc: each service contributed its own units, equipment, and command relationships, the aircrews had not trained together, and there was no standing joint special-operations command to own the mission. The capability had to be improvised, and the improvisation failed. Eagle Claw catalyzed the creation of JSOC, the Goldwater-Nichols reforms, and ultimately U.S. Special Operations Command.

THE FULL CASE, IN FIVE BEATS

Background — No standing joint command existed; the rescue force was drawn ad hoc from four services
What Happened — Three RH-53s failed at Desert One; on withdrawal a helicopter struck a C-130 killing eight
The Investigation — Holloway named ad hoc assembly, untrained aircrews, and a cover story driven aircraft choice
The Capability Gap — Each service was competent in its lane; cross-service integration as a deliverable did not exist
Aftermath & Reform — Reform built JSOC, Goldwater Nichols in 1986, and USSOCOM in 1987 as standing joint capability

KEY REFERENCES

- Holloway Special Operations Review Group, Rescue Mission Report (1980) — the ad-hoc assembly and equipment choices (paraphrased).
- Holloway Commission (1980) — the Desert One sequence and the helicopter–C-130 collision.
- Goldwater-Nichols Department of Defense Reorganization Act of 1986, Pub. L. 99-433.

LENS APPLICABILITY

LENS uses Eagle Claw in LEN 5 as a worked case for cross- organizational capability requirements and in LEN 8 for the institutional reform that followed; it pairs with INPO (Case 16) as the defense analog of failure-driven institution building. The seven-year arc to Goldwater-Nichols (1986) and USSOCOM (1987) is itself a measurement: the institutional response time when the fix requires statutory action.

CASE 47 AVIATION 2005

Helios Airways Flight 522

Shortly after takeoff from Larnaca in August 2005, the crew of Helios 522 misread a warning horn. The same intermittent horn that signals an incorrect takeoff configuration on the ground sounds, in flight, when cabin altitude climbs past about 10,000 feet — and the pressurization selector had been left in "manual" after a leak check the night before. The cabin never pressurized. The crew, troubleshooting the wrong fault, lost consciousness to hypoxia; the Boeing 737 flew on autopilot until it ran out of fuel and crashed into a hillside near Athens, killing all 121 aboard. The investigation found a single cue carried two meanings with no differentiation, and the airline's training had drilled only the ground meaning.

THE FULL CASE, IN FIVE BEATS

Background — A maintenance leak check left the pressurization selector on manual and pre flight checks missed it

What Happened — Cabin failed to pressurize; the crew treated the warning horn as a configuration fault and went hypoxic

The Investigation — Hellenic investigators called the warning misidentification a critical irrecoverable error

The Capability Gap — One horn carried two meanings without differentiation while training drilled only the ground meaning

Aftermath & Reform — Reforms changed Boeing warning logic, pressurization checklists, and pre flight verification training

KEY REFERENCES

- Hellenic Air Accident Investigation & Aviation Safety Board, Aircraft Accident Report 11/2006 — probable cause and the warning misinterpretation (paraphrased).
- AAIASB Report 11/2006 — the manual pressurization selector and failure to pressurize.
- AAIASB Report 11/2006 — the dual-meaning warning horn and the training emphasis.

LENS APPLICABILITY

LENS uses Helios in LEN 5 as a case in cue ambiguity and in LEN 2 as a Human-AI Teaming case where the interface failed to communicate which of two possible system states it was warning about. Students redesign the cue.

CASE 48 AVIATION 1996

AeroPerú Flight 603

While polishing the fuselage of a Boeing 757 in Lima in October 1996, maintenance staff taped over the static ports and failed to remove the tape before flight. With the static system blocked, the aircraft's altimeters, airspeed indicators, and altitude-alert and overspeed warnings all read inconsistently. The crew received a cascade of contradictory warnings — overspeed, stick shaker, ground proximity — and could not determine their true altitude or speed. Believing they were higher than they were, they flew into the Pacific, killing all 70 aboard. The investigation named the maintenance error as the primary cause but stressed that every cockpit instrument depended on the same blocked sensor: the apparent redundancy was an illusion, and the crew's training had no procedure for "everything you see is wrong."

THE FULL CASE, IN FIVE BEATS

Background — Ground crew taped over the static ports during cleaning and the tape was never removed
What Happened — All air data instruments fed false contradictory readings; believing they were higher the crew struck the Pacific
The Investigation — Peruvian investigators found no independent reference since every instrument fed off the blocked source
The Capability Gap — Apparent cockpit redundancy was illusory at the source and the training had assumed the case away
Aftermath & Reform — Industry tightened conspicuous covering and removal verification for static ports and pitot tubes

KEY REFERENCES

- Peru Dirección General de Aeronáutica Civil, accident investigation board, final report on AeroPerú 603 (December 1996; with NTSB/FAA/Boeing participation) — primary cause and the instrument cascade (paraphrased).
- Peru CIAA report (1996) — the static-port tape and the contradictory warnings.
- Peru CIAA report (1996) — the absence of an independent cockpit reference.

LENS APPLICABILITY

LENS uses AeroPerú in LEN 5 to teach the difference between apparent and actual redundancy and in LEN 2 for the human role when all interface data is unreliable. Studio projects examine the "trust nothing" procedure design.

CASE 49 AVIATION 2019

Atlas Air Flight 3591

On approach to Houston in February 2019, the first officer of Atlas Air 3591 — an Amazon Prime Air cargo flight — inadvertently triggered the go-around mode during a turbulent descent. Experiencing a somatogravic illusion that the aircraft was pitching up, he pushed the nose down sharply and dove the Boeing 767 freighter into Trinity Bay, killing all three aboard. The NTSB found the first officer had a long, undisclosed history of training failures across multiple carriers — and that Atlas could not see it, because the Pilot Records Database mandated by Congress after Colgan Air 3407 (Case 43) was not yet operational. Atlas relied on the older PRIA system, which surfaced only five years of history. The case is the live recurrence of Colgan: the information existed; the data-flow system was not yet complete.

THE FULL CASE, IN FIVE BEATS

Background — The first officer's prior training failures were undisclosed and PRIA's window was too short to surface them
What Happened — After an inadvertent go around activation a somatogravic illusion drove a hard push and a dive into Trinity Bay
The Investigation — NTSB cited startle, spatial disorientation, the training pattern, and Atlas's inability to see the full record
The Capability Gap — PRD authorization without full coverage left the same Colgan era information flow gap partly open in 2019
Aftermath & Reform — The PRD final rule took effect for Part 121 in 2022 with full historical coverage phasing in through 2024

KEY REFERENCES

- NTSB, Aircraft Accident Report: Atlas Air Flight 3591, NTSB/AAR-20/02 (2020) — probable cause and contributing factors (paraphrased).
- NTSB/AAR-20/02 (2020) — the inadvertent go-around activation and the dive into Trinity Bay.
- NTSB/AAR-20/02 (2020) — the first officer’s training history and Atlas’s record-access limits.

LENS APPLICABILITY

LENS uses Atlas Air 3591 in LEN 4 as a case for measurement-system coverage (the PRD had been built but its mandatory coverage was incomplete), and in LEN 8 for the iteration cycle of reform: an intervention that leaves an aperture creates a measurable harm inside the aperture.

CASE 50 AVIATION 2015

TransAsia Airways Flight 235

Forty-three seconds after takeoff from Taipei Songshan in February 2015, the right engine of a TransAsia ATR 72 auto-feathered following a sensor fault. Working from memory under acute time pressure, the crew shut down the left engine — the one still producing thrust — leaving the aircraft with no power and too little altitude to recover. It clipped a viaduct, struck a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. The Aviation Safety Council found the captain had failed proficiency checks in the year before the accident and that the crew had not used the published engine-shutdown checklist, responding instead from a flawed memory of the procedure. The wrong-engine error mirrors Kegworth (Case 30) twenty-six years earlier.

THE FULL CASE, IN FIVE BEATS

Background — Captain had failed recent proficiency checks and was flagged for weak abnormal procedure handling

What Happened — Forty three seconds after takeoff the right engine auto feathered and the crew shut down the working left engine

The Investigation — Taiwan ASC found the crew shut down the wrong engine in about fifteen seconds from memory not the checklist

The Capability Gap — Under acute time pressure crews default to memory; incomplete memory drives action away from the checklist

Aftermath & Reform — TransAsia ceased operations in 2016; the case argues the wrong engine pattern is an un engineered intervention

KEY REFERENCES

- Aviation Safety Council (Taiwan), Aviation Occurrence Report: TransAsia Airways Flight GE235, final report (2016) — findings and the wrong-engine shutdown (paraphrased).
- ASC final report (2016) — the autofeather event and the crash sequence.
- ASC final report (2016) — non-use of the checklist and the captain's proficiency record.

LENS APPLICABILITY

LENS uses TransAsia 235 in LEN 5 as a recurrence case for well-known capability problems (wrong-engine shutdown) and in LEN 2 to teach the difference between checklist-driven response and memory-driven response under stress.

CASE 2 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

The Boeing 737 MAX was a re-engined 737 meant to fly without new pilot training — the commercial promise that sold it against the Airbus A320neo. Its bigger engines changed the jet's pitch behavior, so Boeing added software (MCAS) to mask the difference, then kept it out of the manuals and training and let it fire on a single angle-of-attack sensor. When that sensor failed on Lion Air 610 (2018) and Ethiopian 302 (2019), MCAS forced the nose down and crews who had never heard of it could not recover; 346 people died and the fleet was grounded. Investigations found the training omission was deliberate, to avoid certification cost. The MAX is the book's inverse case: human capability engineered out of a system to save the price of sustaining it.

THE FULL CASE, IN FIVE BEATS

Background — Re-engined 737 sold on no new pilot training; MCAS hid the handling change
What Happened — Single-sensor MCAS forced two jets down, killing 346 across both crashes
The Investigation — Internal records show training omission was deliberate; certification was largely self-delegated
The Capability Gap — Training absence was a contract deliverable, not an oversight; pilots engineered out
Aftermath & Reform — Twenty-month grounding; MCAS redesigned, simulator training required, FAA oversight tightened

KEY REFERENCES

- U.S. House Committee on Transportation and Infrastructure, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (Final Committee Report, Sept. 2020) — the MCAS omission, the 2013 minutes, the 2016 survey, and the "diminished safety" conclusion.
- U.S. House report (2020) and MCAS design summary — MCAS firing on a single angle-of-attack sensor; the Lion Air 610 (189 killed) and Ethiopian 302 (157 killed) accident sequences.
- U.S. House Transportation Committee report (2020) — Boeing internal communications and the certification-and-training-impact reasoning (quoted).

LENS APPLICABILITY

LENS treats capability requirements as system requirements: derived from operational analysis, traceable to design decisions, falsifiable against operational outcomes. A LENS-trained engineer in the MAX certification process would have produced a capability-requirements artifact mapping pilot tasks to system behaviors — and flagged the proposed manual omission as an unmitigated requirements gap.

CASE 7 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

The Therac-25, a radiation-therapy machine, massively overdosed at least six patients between 1985 and 1987, killing at least three. Its predecessors used hardware interlocks to stop the high-energy beam from firing without its target in place; to save cost, the Therac-25 removed them and trusted software — adapted from the older machines and never engineered for safety — to keep the beam modes separated. A race condition let a fast operator trigger the full beam with no target, while the console showed only a meaningless “Malfunction 54.” Leveson and Turner’s 1993 investigation, the founding case of software-safety engineering, found a systemic failure: a safeguard removed with nothing put in its place. The lesson — a safeguard you delete does not remove the hazard; it relocates it to whatever you failed to build.

THE FULL CASE, IN FIVE BEATS

Background — Hardware interlocks removed to save cost; safety case migrated silently into software
What Happened — Race condition let fast operators fire full beam with no target
The Investigation — Manufacturer denied harm; Leveson and Turner found systemic, not single-bug, failure
The Capability Gap — Interlock was the safety case; nothing took its load when removed
Aftermath & Reform — Founded software-safety engineering; deleted safeguards relocate hazard to whatever replaces them

KEY REFERENCES

- N. G. Leveson & C. S. Turner, “An Investigation of the Therac-25 Accidents,” IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software.
- Leveson & Turner (1993) — the race condition, the uninformative “Malfunction 54”, overdoses up to 100×, six accidents, and at least three deaths.
- Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer’s denial and the systemic findings.

LENS APPLICABILITY

LENS frames Therac-25 as a **Design-Out** failure made visible through Interface and Governance pathologies. Studio projects in LEN 5 ask students to produce capability-load diagrams tracing every safety function to a hardware backstop, a software check, or a trained operator action with the information needed to perform it. LEN 7 examines incident reporting as governance.

CASE 19 DEFENSE 1991

Patriot Missile / Dhahran

On 25 February 1991 a Scud missile struck a U.S. barracks in Dhahran, Saudi Arabia, killing 28 soldiers and wounding about a hundred; the Patriot battery defending the area never engaged it. Designed for short engagements against Soviet aircraft in Europe, the Patriot tracked time in a register whose tiny rounding error grew with every hour of uptime. After about a hundred hours of continuous operation the Gulf War demanded, the radar's range gate was off by a third of a second — enough to look in the wrong place. Israeli operators had flagged the drift two weeks earlier, and a patch reached Dhahran the day after the strike. The capability to hold accuracy under sustained use was assumed away at design time, and no one carried that assumption forward when the mission changed.

THE FULL CASE, IN FIVE BEATS

Background — Built for short European engagements; Gulf War demanded continuous fixed-site operation
What Happened — Fixed-point time error grew with uptime; Scud passed unengaged, 28 killed
The Investigation — Israeli warning preceded strike by weeks; patch arrived one day late
The Capability Gap — Original concept did not require sustained accuracy; assumption never traveled with redeployment
Aftermath & Reform — Patch distributed, reboot intervals defined; canonical lesson about assumptions outliving their context

KEY REFERENCES

- U.S. General Accounting Office, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia, GAO/IMTEC-92-26 (1992) — the design-context mismatch and continuous-operation use.
- R. Skeel, "Roundoff Error and the Patriot Missile," SIAM News 25(4): 11 (1992) — the 24-bit fixed-point time truncation and the 0.34-second range-gate drift after 100 hours.
- GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed.

LENS APPLICABILITY

LENS treats Patriot as the textbook example of **Capability Degradation Under System Change**. LEN 5 methods require operators of any transitioning system to have current capability-relevant documentation specifying what assumptions of the original design have changed. LEN 8 examines how organizational knowledge about design constraints travels from engineering to operations — and why, here, it arrived a day late.

CASE 51 INDUSTRIAL 1971 – 1978

Ford Pinto Fuel Tank

The Ford Pinto, a rushed early-1970s subcompact, placed its fuel tank where rear-impact collisions could rupture it; fuel-fed fires killed and burned people, and in Grimshaw v. Ford (1981) a jury found Ford liable, having known of the hazard and not fixed it. A **common misconception**, popularized by a 1977 Mother Jones exposé, holds that Ford ran a cost-benefit calculation and chose to pay death settlements rather than install an $11 fix. The memo invoked for that story was actually a generic federal submission about rollover fires across the whole U.S. fleet — it never mentioned the Pinto. Strip away the myth and the real designed-out failure remains: rear-impact survivability lost to cost and packaging, and a known hazard went unfixed. It is the founding engineering-ethics case — and a lesson in accuracy.

THE FULL CASE, IN FIVE BEATS

Background — Rushed subcompact with tank behind rear axle and little crush space, known to engineers.
What Happened — Rear impacts produced fuel-fed fires; Grimshaw verdict and 1.5 million-vehicle recall followed.
The Investigation — Mother Jones cost-benefit story rested on a generic rollover memo, not a Pinto ledger.
The Capability Gap — Real failure was rear-impact survivability quietly traded against cost, weight, and packaging.
Aftermath & Reform — Founding ethics case shaped fuel-system rules and taught how myths obscure lessons.

KEY REFERENCES

- M. Dowie, “Pinto Madness,” Mother Jones (Sept./Oct. 1977) — the contemporaneous exposé that popularized the $11-fix / cost-benefit framing.
- Grimshaw v. Ford Motor Co., 119 Cal. App. 3d 757 (1981) — Ford held liable for the rear-impact fuel-system hazard; punitive damages upheld in reduced amount.
- G. T. Schwartz, “The Myth of the Ford Pinto Case,” Rutgers Law Review 43: 1013–1068 (1991) — the Grush-Saunby memo was a generic NHTSA rollover-fire submission for the whole fleet, not a Pinto-specific litigation calculation; the death toll was far lower than popularly claimed. (See AUDIT: corrects the prior edition’s framing.)

LENS APPLICABILITY

LENS uses the Pinto in LEN 7 as the foundational engineering-ethics case and in LEN 1 as a problem-framing case for what happens when safety capability is treated as a cost line item.

CASE 52 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

Takata built its airbag inflators around ammonium nitrate — cheap and energetic but unstable, and used by almost no one else without a moisture-absorbing desiccant. Takata's inflators omitted the desiccant. Over years of heat and humidity the propellant degraded and could rupture the metal housing on deployment, spraying shrapnel at the driver; more than two dozen deaths and hundreds of injuries followed. Takata's own tests had shown the ruptures, but it reported them as isolated anomalies and, in places, falsified data; in 2017 it pleaded guilty to wire fraud, paid roughly $1 billion, and went bankrupt. The recall — over 100 million inflators across 19 automakers — is the largest in automotive history. Two capabilities were designed out: the desiccant, and the independent verification regulators never had.

THE FULL CASE, IN FIVE BEATS

Background — Inflators built around cheap, unstable ammonium nitrate without the desiccant competitors added.
What Happened — Heat and humidity degraded the propellant; ruptures sprayed shrapnel and killed drivers.
The Investigation — Takata reported ruptures as isolated anomalies, manipulated test data, and pleaded guilty to fraud.
The Capability Gap — Designed-out capabilities were the desiccant and the regulator's independent verification pipeline.
Aftermath & Reform — Recall outlived the bankrupt firm and pushed regulators toward coordinated replacement management.

KEY REFERENCES

- U.S. National Highway Traffic Safety Administration, Takata air-bag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism.
- NHTSA recall record and investigative reporting (Reuters, New York Times) — 100M+ inflators across 19 automakers; the largest automotive recall in history; deaths and injuries from ruptures.
- U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata's subsequent bankruptcy.

LENS APPLICABILITY

LENS uses Takata in LEN 4 as a measurement-system case (the manufacturer's data was reported but not interpreted as a class) and in LEN 7 as an industrial-governance failure spanning manufacturer, regulator, and customer auditors.

CASE 53 INDUSTRIAL 2002 – 2014

GM Ignition Switch

A faulty ignition switch in several GM compact cars (the Chevrolet Cobalt, Saturn Ion) could slip from “run” to “accessory” while driving, cutting power steering and brakes and — fatally — disarming the airbags. GM engineers identified it in 2002. In 2006 an engineer approved a redesigned switch but did not change its part number, so the fix propagated as neither a revision nor a recall, and defective cars kept selling. The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through GM’s compensation fund. The Valukas report found a culture that absorbed safety concerns, and GM paid a $900 million federal penalty. What was designed out was not a part but the pathway by which a known safety problem reaches the decision to recall.

THE FULL CASE, IN FIVE BEATS

Background — Compact-car switches lacked detent torque; jostles cut power and disarmed airbags.
What Happened — Engineer approved a redesigned switch in 2006 without changing its part number.
The Investigation — Valukas found the GM nod, the GM salute, and failure to use escalation processes.
The Capability Gap — Designed out was the pathway from a known fix to the recall decision.
Aftermath & Reform — Barra restructured safety governance and the case became a teaching example in psychological safety.

KEY REFERENCES

- A. R. Valukas, Report to the Board of Directors of General Motors Company Regarding Ignition Switch Recalls (Jenner & Block, 2014) — the 2002 identification of the defect and the switch-torque mechanism.
- U.S. NHTSA investigation of the GM ignition switch (2014) and the GM recall record — 2.6 million vehicles; 124 deaths via the GM compensation fund.
- Valukas Report (2014) — the unchanged part number, the “GM nod” and “GM salute,” and the failure to use established escalation processes (quoted).

LENS APPLICABILITY

LENS uses GM in LEN 7 as a corporate-governance failure case, in LEN 8 to study Valukas-style retrospective institutional analysis, and in LEN 4 for the part-number-as-measurement-instrument story (changing the part without changing the number suppressed the signal).

CASE 54 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

The Mars Climate Orbiter, a NASA spacecraft built by Lockheed Martin and navigated by JPL, was lost on arrival at Mars in September 1999. Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary between the two teams, so every trajectory correction was mis-modeled and the error accumulated over the cruise. The orbiter arrived too low, into the atmosphere, and burned up; the orbiter and its $125 million were lost to a unit mismatch. The investigation found no individual blunder — both contractors did their own work correctly. What failed was the interface between them, which had no owner, specification, or verification step. It is the canonical case of interface-as-requirement.

THE FULL CASE, IN FIVE BEATS

Background — Faster, better, cheaper mission split between Lockheed building and JPL navigating the orbiter.
What Happened — Ground software reported pound-force seconds while navigation expected newton-seconds; orbiter burned up at Mars.
The Investigation — Board found no individual blunder; the unverified boundary between two correct halves failed.
The Capability Gap — Missing capability was an owned, specified, verified interface between the two organizations.
Aftermath & Reform — Loss tightened interface management and became the canonical software case of interface-as-requirement.

KEY REFERENCES

- NASA, Mars Climate Orbiter Mishap Investigation Board: Phase I Report (Nov. 1999) — mission, the Lockheed/JPL split, and the program context.
- NASA MCO MIB Report (1999) — the pound-force-second vs newton-second mismatch (factor 4.45), the accumulated navigation error, and atmospheric destruction on 23 Sept. 1999 (root-cause statement quoted).
- NASA MCO MIB Report (1999) — the missing verified interface specification, the absent end-to-end validation, and the unresolved cruise-trajectory anomalies.

LENS APPLICABILITY

LENS uses Mars Climate Orbiter in LEN 5 to teach interface-as- requirement and in LEN 8 to discuss cross-contractor capability boundaries. The case is the foundational software-engineering parallel to Patriot (Case 19).

CASE 55 TECHNOLOGY 2012

Knight Capital Trading Loss

On 1 August 2012 Knight Capital, a major U.S. market maker, deployed new order-routing software to seven of its eight servers and missed the eighth. The new code reused a flag that, on the eighth server's old software, reactivated long-dead "Power Peg" test code never removed from the repository. At the opening bell it fired millions of unintended orders; in about 45 minutes Knight lost roughly $440 million — more than the firm was worth — and was effectively acquired within months. The SEC found Knight had no procedure to verify the deployment across all servers and no controls to halt the runaway orders. The capability designed out was deployment verification; the dead code was technical debt that exercised its option at the worst possible moment.

THE FULL CASE, IN FIVE BEATS

Background — Major market maker prepared routine update for the NYSE Retail Liquidity Program launch.
What Happened — Eighth server missed the update; a reused flag woke dormant Power Peg code.
The Investigation — SEC found no deployment verification, no consistency check, and no controls to halt orders.
The Capability Gap — Designed-out capability was deployment verification; dead code was a standing option on failure.
Aftermath & Reform — Case became canonical for deployment as engineering deliverable and sharpened market-access controls.

KEY REFERENCES

- U.S. Securities and Exchange Commission, Order Instituting Administrative and Cease-and-Desist Proceedings, In re Knight Capital Americas LLC (2013) — the firm, the Retail Liquidity Program launch, and the deployment.
- SEC Order (2013) and Knight Capital 8-K filing (2012) — the missed eighth server, the reactivation of the dormant "Power Peg" code, the $440 million loss in 45 minutes, and the near-collapse.
- SEC Order (2013) — the absence of a second-technician deployment verification, the lack of an automated code-consistency check, inadequate order controls, and the $12 million penalty (quoted).

LENS APPLICABILITY

LENS uses Knight Capital in LEN 5 to teach deployment-as-capability and in LEN 9 for the technical-debt argument: every line of dead code carries an option on a future failure. Students design the deployment deliverable that would have caught the eighth server.

CASE 4 ENERGY 2010

Deepwater Horizon

On 20 April 2010 the Macondo well blew out beneath the Deepwater Horizon rig in the Gulf of Mexico, killing 11 workers and releasing roughly 4.9 million barrels of oil — the largest marine spill in U.S. history, and more than $65 billion in eventual costs to BP. The crew had misread the negative-pressure test, the primary check of well integrity, and kept working a well that was no longer sealed. The National Commission found human error, not equipment, the root cause, and judged the failures so systematic they cast doubt on the safety culture of the whole industry. Every defense — procedure, training, supervision, contractor coordination, equipment — had drifted independently until none caught the others. It is the book's canonical multi-layer normalization failure.

THE FULL CASE, IN FIVE BEATS

Background — Macondo blowout killed eleven and unleashed the largest U.S. marine oil spill
What Happened — Crew misread the negative-pressure test, accepted a wrong explanation, and displaced the mud
The Investigation — Commission found human error so systematic it indicted the whole industry's safety culture
The Capability Gap — Procedure, training, supervision, contractors, and equipment had drifted independently until none caught the others
Aftermath & Reform — Regulator was split, well-integrity rules tightened, and the multi-layer drift named as the deeper lesson

KEY REFERENCES

- National Commission on the BP Deepwater Horizon Oil Spill, Deep Water: The Gulf Oil Disaster and the Future of Offshore Drilling (Report to the President, 2011) — human error as root cause; the misread negative-pressure test.
- National Commission (2011) — the well-control sequence and mud-displacement decision.
- National Commission (2011) — "systematic failures in risk management… place in doubt the safety culture of the entire industry" (quoted).

LENS APPLICABILITY

LENS uses Deepwater Horizon in LEN 5 as the canonical case for multi-layer capability analysis. Students reconstruct each defense layer, identify the deviation accumulated in each, and design the measurement system that would have surfaced the drift before April 20, 2010. LEN 8 examines contractor-coordination as a capability boundary problem.

CASE 6 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

NASA lost two Space Shuttle crews to the same organizational pathology seventeen years apart: Challenger in 1986, when O-ring seals failed in cold weather, and Columbia in 2003, when foam debris breached the wing's thermal protection — fourteen astronauts in all. Both flaws had been seen repeatedly and accepted as routine. Sociologist Diane Vaughan named the mechanism "normalization of deviance" from the Challenger investigation; the Columbia Accident Investigation Board found the same culture intact and concluded NASA's organizational culture had as much to do with the accident as the foam. The pair is the strongest single-institution evidence that culture is an engineerable property of a system — and that a pathology diagnosed but left unrepaired will recur, at the same cost.

THE FULL CASE, IN FIVE BEATS

Background — Shuttle flew with known O-ring and foam flaws reclassified flight by flight as routine
What Happened — Challenger broke apart on a cold morning; Columbia disintegrated on reentry seventeen years later
The Investigation — Both boards found the same suppressed safety concerns and the same culture intact
The Capability Gap — Vaughan named normalization of deviance; the diagnosis sat on record without engineered remediation
Aftermath & Reform — Reforms followed each loss, yet the cultural mechanism stayed unrepaired until the Shuttle retired

KEY REFERENCES

- Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986) — the O-ring failure and the cold-weather launch decision.
- Columbia Accident Investigation Board, Report Vol. I (2003) — the foam strike and the recurrence of the cultural pattern.
- CAIB (2003) — "the NASA organizational culture had as much to do with this accident as the foam" (quoted).

LENS APPLICABILITY

LENS uses the pair in LEN 1 to introduce normalization of deviance as a systems concept, in LEN 8 to examine why organizational learning failed across a 17-year interval, and in LEN 7 to address the governance failure that allowed a known pathology to persist.

CASE 11 DEFENSE 1991 – PRESENT

V-22 Osprey

The V-22 Osprey — the tiltrotor flown by the Marines, Air Force, and Navy — has had about 17 hull-loss accidents and roughly 65 fatalities since 1991, including 19 Marines in a single 2000 test crash and 8 airmen off Yakushima, Japan, in 2023. The Yakushima crash traced to cracks in a transmission gear (a flaw in the X-53 steel alloy) and a pilot's decision to keep flying through warnings. In December 2025, GAO and a NAVAIR review found the V-22's joint program office had failed for years to assess and address mounting safety risks, that serious-mishap rates exceeded comparable fleets, and that some fixes stretch toward 2034. The Osprey is the steady-state form of normalization of deviance:

a documented, reviewed shortfall accepted as the cost of flying the airframe.

THE FULL CASE, IN FIVE BEATS

Background — Tiltrotor shared awkwardly across three services has logged seventeen hull losses since 1991

What Happened — Yakushima crash killed eight after gear cracks and a pilot pressing through warnings

The Investigation — GAO and NAVAIR found years of unaddressed risk, elevated mishap rates, eighteen-year gearbox lag

The Capability Gap — Documented shortfall persists because parallel service adjustments never converge on resolution

Aftermath & Reform — Groundings and redesigns continue while full gearbox fixes stretch toward 2034

KEY REFERENCES

- Compiled V-22 accident record — 17 hull losses and 65 fatalities since 1991, including the 2000 Marana test crash (19 Marines).
- U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings.
- U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office's failure to assess and address risks.

LENS APPLICABILITY

LENS treats V-22 in LEN 5 as a multi-service capability-coordination problem and in LEN 8 as a study in long-cycle organizational learning failure. Students design the evidence system that would distinguish a true reduction in mishap rate from the natural variation of a chronically marginal platform.

CASE 56 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

On 23 March 2005 a startup at BP's Texas City refinery overfilled a distillation tower — operators were working from instruments that had malfunctioned for years — and venting hydrocarbon vapor found an ignition source and exploded, killing 15 workers in trailers parked beside the unit and injuring 180. The Chemical Safety Board found accumulated safety-culture decay, deferred maintenance, and a celebrated cost-cutting program. Its central finding reshaped U.S. industrial safety: BP's **personal**-safety metrics were among the best in the industry, while its **process**-safety state — the integrity of the hazardous process itself — had been drifting, unmeasured. Texas City is the canonical evidence that the wrong measurement, reported as a win, is worse than no measurement at all.

THE FULL CASE, IN FIVE BEATS

Background — Deferred maintenance and degraded instruments tolerated; personal-injury rates among the industry's best

What Happened — Startup overfilled tower past safe level; vented vapor ignited, killing fifteen workers in trailers

The Investigation — CSB drew the distinction reshaping the field; personal-safety indicators are not process-safety indicators

The Capability Gap — Excellent injury record made process-safety drift harder to see; wrong measurement worse than none

Aftermath & Reform — Baker Panel followed; process-safety distinction entered mainstream regulation; count what can kill you

KEY REFERENCES

- U.S. Chemical Safety and Hazard Investigation Board, Refinery Explosion and Fire, Investigation Report 2005-04-I-TX (2007) — the startup, malfunctioning instruments, and 15 killed / 180 injured.
- CSB (2007) — accumulated safety-culture decay, deferred maintenance, and the siting of occupied trailers beside a hazardous unit.
- CSB (2007) — "indicators of personal safety are not indicators of process safety" (quoted).

LENS APPLICABILITY

LENS uses Texas City in LEN 4 as the canonical wrong-measurement case and in LEN 7 for the governance failure that allowed years of cost-cutting to be reported as wins. Studio projects design the process-safety measurement deliverable.

CASE 57 ENERGY 2002

Davis-Besse Reactor Head Corrosion

During a 2002 refueling outage at the Davis-Besse nuclear plant in Ohio, workers found a football-sized cavity eaten through six inches of the carbon-steel reactor-vessel head by leaking boric acid, leaving only a thin layer of stainless cladding between the cavity and reactor coolant at 2,200 psi — a serious near-miss for a loss-of-coolant accident. The leakage had been visible for years and inspections deferred; FirstEnergy had won an NRC deferral of a mandatory inspection to fit its refueling schedule, and the NRC's Inspector General later found the decision had weighted economics over safety. Davis-Besse is the canonical post-Three-Mile-Island near-miss — evidence that even an industry that built INPO to engineer safety can let regulator-operator dynamics erode it.

THE FULL CASE, IN FIVE BEATS

Background — Post-TMI U.S. nuclear regime knew the boric-acid corrosion hazard across the reactor fleet

What Happened — Refueling outage revealed football-sized cavity through the head; only thin cladding held back coolant

The Investigation — FirstEnergy won an NRC inspection deferral; OIG found economics weighted over safety

The Capability Gap — Plant engineering was sound; independence of the oversight layer above operations had quietly eroded

Aftermath & Reform — INPO and NRC tightened head-inspection requirements; institutional capability erodes if not re-engineered

KEY REFERENCES

- U.S. NRC Office of Inspector General, NRC's Regulation of Davis-Besse Regarding Damage to the Reactor Vessel Head (Case No. 02-03S, December 2002) — the post-TMI regulatory regime and oversight failures.
- NRC event records and OIG (2002) — the 6-inch corrosion cavity, the remaining cladding, and the 2,200-psi coolant margin.
- NRC Bulletin 2001-01 and the FirstEnergy inspection-deferral decision aligned to the February 2002 outage.

LENS APPLICABILITY

LENS uses Davis-Besse in LEN 7 to study regulatory-capture dynamics and in LEN 8 to compare with INPO's earlier success. The case is a reminder that institutional capability requires sustained re-engineering.

CASE 58 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

Between roughly 2005 and 2009, Stafford Hospital, run by the Mid Staffordshire NHS Foundation Trust, subjected patients to appalling neglect — left without food, water, or basic care — while mortality ran above expected. The trust had been chasing financial targets that depended on staffing cuts. Robert Francis QC's public inquiry produced a 2,000-page report and 290 recommendations, identifying the structural problem as the gap between the trust's reported performance and patients' actual experience: every governance layer above the ward received reports that targets were being met, and none checked those reports against what was happening to patients. Mid Staffs is the dataset's strongest case for the harm done when measurement and reality diverge over years.

THE FULL CASE, IN FIVE BEATS

Background — Pursuing Foundation Trust status, the board cut ward staffing while reported targets stayed met

What Happened — Patients neglected for years in appalling conditions; mortality ran substantially above expected

The Investigation — Francis Inquiry produced 2,000 pages and 290 recommendations; system as a whole failed

The Capability Gap — Like the VA case, every governance layer acted on metrics; no one checked against patients

Aftermath & Reform — Berwick review pressed for learning over targets; reporting can run clean while reality fails

KEY REFERENCES

- R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets.
- Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality.
- Francis QC (2013) — the 2,000-page report and 290 recommendations.

LENS APPLICABILITY

LENS uses Mid Staffs in LEN 4 for the divergence-of-measurement problem and in LEN 7 for the governance failure across multiple layers. Studio projects examine the Francis recommendations as a template for institutional reform deliverables.

CASE 59 ENERGY INDUSTRIAL 2006

Sago Mine Disaster

On 2 January 2006 lightning ignited methane in a sealed area of the Sago Mine in West Virginia; the seals failed, the explosion propagated, and thirteen miners were trapped behind it. Twelve died of carbon-monoxide poisoning over the hours that followed; only one, Randal McCloy Jr., survived. A communications failure briefly told the nation — and the families — that twelve had been found alive, when the opposite was true. The MSHA investigation found seals built to an inadequate design, an inadequate emergency plan, and lapsed self-rescue training — each marginal for years, all inadequate together in the one window that mattered. Sago drove the MINER Act of 2006, strengthening mine-rescue requirements and mandating underground refuge chambers.

THE FULL CASE, IN FIVE BEATS

Background — Seals, emergency plan, and self-rescue training were each marginally adequate, never tested at edge

What Happened — Lightning ignited methane; seals failed; twelve miners died of carbon-monoxide poisoning over hours

The Investigation — MSHA found inadequate seal design, plan, and lapsed training; combined shortcomings proved lethal together

The Capability Gap — No single failure was decisive; marginal-everywhere is itself a system-level hazard

Aftermath & Reform — MINER Act strengthened rescue requirements; mandated refuge chambers; reform addressed the combination, not one cause

KEY REFERENCES

- U.S. Mine Safety and Health Administration, Report of Investigation: Sago Mine (2007) — the seal design, emergency plan, and self-rescue training.
- MSHA (2007) — the explosion sequence: lightning ignition in the sealed area, seal failure, twelve dead and one survivor.
- MSHA (2007) — the inadequate seal design, emergency plan, and lapsed self-rescue training.

LENS APPLICABILITY

LENS uses Sago in LEN 5 for cumulative-inadequacy analysis and in LEN 8 for the legislative-reform arc that followed. Studio projects compare Sago and Upper Big Branch (Case 60) as paired cases.

CASE 60 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

On 5 April 2010 coal dust and methane ignited at Massey Energy's Upper Big Branch mine in West Virginia, killing 29 miners — the worst U.S. mine disaster in forty years. Investigators found Massey had kept two sets of records: an internal log of actual conditions and a separate, clean log for federal inspectors. Foremen were directed to suppress methane readings, disable monitors, and falsify pre-shift examinations — not as an aberration but as a stable routine across years and management layers. CEO Don Blankenship was convicted of a misdemeanor conspiracy to violate mine-safety standards — the first U.S. mining CEO criminally convicted on such a charge. Upper Big Branch is the dataset's clearest case of a measurement system engineered deliberately to defeat the regulator.

THE FULL CASE, IN FIVE BEATS

Background — Massey ran two sets of records; the clean log disabled the regulator's mechanism of sight

What Happened — Coal dust and methane ignited, killing twenty-nine; the sanitized records hid the conditions

The Investigation — MSHA and DOJ found suppressed readings and disabled monitors; CEO Blankenship convicted of misdemeanor conspiracy

The Capability Gap — Dual records were stable institutional practice spanning years; measurement engineered as deception

Aftermath & Reform — Conviction set accountability marker; unanswered is what independent signal could expose divergence live

KEY REFERENCES

- U.S. MSHA, Internal Review of MSHA's Actions at Upper Big Branch and the accident investigation (2011–2012) — the dual records and the 29 deaths.
- Governor's Independent Investigation Panel (J. McAtee, 2011) — mine conditions: methane, ventilation, and coal-dust inerting.
- MSHA and U.S. DOJ findings — suppressed methane readings, disabled monitors, and falsified pre-shift examinations.

LENS APPLICABILITY

LENS uses Upper Big Branch in LEN 4 as the most explicit case for measurement-system protection and in LEN 7 as a corporate-criminal accountability case. Studio projects examine what regulator-side architecture would have detected the dual-records system earlier.

CASE 61 ENERGY 2011

Fukushima Daiichi

On 11 March 2011 the Tōhoku earthquake and the tsunami it spawned overwhelmed TEPCO's Fukushima Daiichi plant: the wave topped the seawall, flooded the emergency diesel generators, and cut cooling to the reactors. Three of the six cores melted down and hydrogen explosions spread radioactive material, displacing some 160,000 people; cleanup is projected above $200 billion. The independent Diet commission (NAIIC), chaired by Kiyoshi Kurokawa, called it a disaster "made in Japan" — the product of regulatory capture and a culture reluctant to challenge utility assumptions; evidence of large historical tsunamis had been discussed internally but never forced a change to the seawall. Other reviews stressed under-estimated external hazards. Fukushima is the post-TMI evidence that safety institutions like INPO must be deliberately built, not assumed.

THE FULL CASE, IN FIVE BEATS

Background — Internal TEPCO assessments discussed larger historical waves; evidence never forced a higher seawall
What Happened — Tōhoku tsunami topped the seawall; inundated generators; three cores melted down, displacing thousands
The Investigation — Kurokawa's NAIIC called it made in Japan; Hatamura and IAEA emphasized under-estimated external hazards
The Capability Gap — Internal hazard evidence existed; Japan lacked an INPO-equivalent with independence to act on it
Aftermath & Reform — New independent Nuclear Regulation Authority created; capability institutions must be deliberately built, not assumed

KEY REFERENCES

- National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission (NAIIC; K. Kurokawa, chair), Report (2012) — the internal tsunami evidence and the regulatory-capture finding.
- NAIIC (2012) — the accident sequence: seawall overtopping, generator inundation, and three core meltdowns.
- NAIIC (2012) — the "made in Japan" cultural and regulatory-capture conclusion (quoted).

LENS APPLICABILITY

LENS uses Fukushima in LEN 8 as the cross-cultural comparison to INPO and in LEN 7 for the regulator-utility dynamics study. The case is paired with INPO and Davis-Besse to triangulate what sustained nuclear-safety capability requires.

CASE 62 ENERGY 2003

Northeast Blackout

On 14 August 2003 a high-voltage transmission line in Ohio sagged into a tree and tripped — an event the grid should have absorbed. But FirstEnergy's control-room alarm system had been silently failing for over an hour, so operators did not know the line was gone. Further lines tripped, and a cascade swept across the Eastern Interconnection; within minutes 50 million people across eight U.S. states and Ontario lost power, at a cost above $6 billion. The U.S.-Canada task force found FirstEnergy lacked situational awareness, its alarm system had failed without notice, vegetation management was poor, and the regional co-ordinator could not intervene. The reforms made reliability standards mandatory and enforceable (FERC Order 693). The gap sat at the automation-operator boundary: a silent failure left operators blind.

THE FULL CASE, IN FIVE BEATS

Background — Eastern Interconnection rides through single-line loss; FirstEnergy's alarm system was silently failing for an hour

What Happened — An Ohio line sagged into a tree; silent alarms left operators blind; cascade blacked out fifty million

The Investigation — Task Force found inadequate situational awareness, vegetation lapses, weak coordinator authority; FERC Order 693 followed

The Capability Gap — Missing capability was the meta-monitor; silence is indistinguishable from a healthy, quiet grid

Aftermath & Reform — New mandatory reliability regime backed by audits and penalties; reliability is a deliverable, not best practice

KEY REFERENCES

- U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade.
- Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence.
- Task Force (2004) — FirstEnergy "did not have adequate situational awareness," plus the vegetation-management and reliability-coordinator findings (quoted).

LENS APPLICABILITY

LENS uses the 2003 blackout in LEN 2 as a Human-AI Teaming case for silent-automation-failure handling and in LEN 8 for the legislative-reform arc that produced enforceable reliability standards.

CASE 18 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

On 3 July 1988, during a surface skirmish with Iranian gunboats in the Persian Gulf, the USS Vincennes shot down Iran Air Flight 655 — a civilian Airbus A300 climbing on a scheduled route — killing all 290 aboard, the deadliest airliner shootdown by a military force. The Aegis combat system worked: its radar correctly showed the aircraft ascending. Yet operators in the Combat Information Center, primed to expect a hostile attack, read the contact as a descending F-14 and fired. The Navy's inquiry attributed the tragedy to human error under extreme stress — confirmation bias and the unconscious distortion of data — not equipment failure. Vincennes is the book's foundational human-AI-teaming case: the most advanced system afloat failed because its interface did not support correct performance under combat stress.

THE FULL CASE, IN FIVE BEATS

Background — Aegis cruiser fought Iranian gunboats near a dual-use airfield in the crowded Persian Gulf

What Happened — Crew misread an ascending Airbus as a diving F-14 and fired; all 290 died

The Investigation — Aegis radar was correct; multiple operators independently saw descent under presumed-hostile framing

The Capability Gap — Interface left burden of overriding expectation to operators stripped of time by combat

Aftermath & Reform — Naval retrospective reframed loss as predictable teaming failure of unguarded decision aids

KEY REFERENCES

- Rear Adm. W. Fogarty, Formal Investigation into the Circumstances Surrounding the Downing of Iran Air Flight 655 (U.S. Navy, 1988) — the engagement and the 290 deaths.
- Fogarty report (1988) — the Aegis system functioned; the aircraft was ascending while the crew perceived a descent.
- Fogarty report (1988) — "human error under extreme stress," confirmation bias, and "unconscious distortion of data" (quoted).

LENS APPLICABILITY

LENS uses Vincennes in LEN 2 as the foundational case for human-AI teaming under operational stress, and in LEN 5 to teach the analysis of capability requirements in conditions the original interface designers never expected. The case sits at the center of the program's argument that interface design is a capability deliverable, not an aesthetic one.

CASE 25 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

The 2009 HITECH Act poured roughly $30 billion into accelerating electronic health records, and adoption surged — but the systems had been built to billing and administrative specifications, not clinical workflow. New categories of harm followed. A disputed but canonical 2005 study found that deploying a commercial order-entry (CPOE) system at one pediatric ICU was associated with a near-doubling of mortality; later deployments showed mixed or improved results, yet the warning stuck. A 2023 survey across 200-plus hospitals found EHR usability is clinicians' top complaint, and that usability scores track patient-safety outcomes. Specific harms recur — a missed abnormal result hidden behind a default, a fatal overdose an unactivated alert would have caught. There is still no regulatory framework monitoring EHR safety. It is the book's live interface-mismatch case.

THE FULL CASE, IN FIVE BEATS

Background — HITECH poured thirty billion into EHRs built to billing specifications, not clinical workflow

What Happened — Disputed 2005 CPOE study tied pediatric ICU mortality to deployment; default and alert harms recurred

The Investigation — A 2023 KLAS survey across hundreds of hospitals tied usability scores to patient safety outcomes

The Capability Gap — Systems work for billing because that was the specification; clinical workflow was never engineered

Aftermath & Reform — Usability guidance has matured but no regulatory regime monitors deployed EHR safety at scale

KEY REFERENCES

- Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program.
- Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result.
- KLAS Arch Collaborative, EHR usability and safety surveys (2023) — usability as the top clinician complaint, correlated with safety outcomes.

LENS APPLICABILITY

LENS treats EHR/CPOE in LEN 7 as the live, ongoing example of Design-Out and Interface failure under governance opacity, in LEN 2 as a human-AI teaming problem (alert fatigue, default surfacing), and in LEN 9 as a computational systems problem (the alerting architecture itself is a learnable model). The Han 2005 pediatric ICU finding is the durable warning; the KLAS 2023 surveys across two hundred hospitals are the contemporary, longitudinal evidence that usability is now itself a patient-safety variable.

CASE 29 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

On the night of 18 March 2018 a modified Volvo running Uber's self-driving system struck and killed Elaine Herzberg as she crossed a road in Tempe, Arizona — the first pedestrian killed by an autonomous vehicle. The NTSB found the safety operator had been watching a video on her phone, but placed heavy blame on Uber: it had not recognized the risk of automation complacency, trained for it, or enforced its own no-phone policy. The system itself was programmed not to brake when a crash was deemed unavoidable, and could not classify a pedestrian who was not near a crosswalk. The human was kept in the loop as a passive monitor — a role the NTSB noted is chronically unperformable — with no infrastructure to make it work. Uber ATG is the book's defining human-AI-teaming case.

THE FULL CASE, IN FIVE BEATS

Background — Uber ATG tested self-driving cars using safety operators as passive surveillance of rare failures
What Happened — A Volvo killed Elaine Herzberg in Tempe; the operator was watching a phone video
The Investigation — NTSB faulted Uber for ignoring automation complacency; emergency braking was suppressed and pedestrian classification limited
The Capability Gap — The monitor role lacked interface, training, and authority; design assumed failure would never arrive
Aftermath & Reform — Uber exited self-driving; industry shifted toward two-operator teams and driver-monitoring systems

KEY REFERENCES

- NTSB, Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian, Highway Accident Report HAR-19/03 (2019) — the Tempe collision and probable cause.
- NTSB HAR-19/03 (2019) — the safety operator's distraction (watching a video) before impact.
- NTSB HAR-19/03 (2019) — Uber "did not adequately recognize the risk of automation complacency"; training and policy failures (quoted in part).

LENS APPLICABILITY

LENS treats this case in LEN 2 as the live exemplar of monitoring as an unsupportable role. Students reconstruct the capability requirements for the safety operator and design the interface, training, and authority structure that would have made the role performable — or made the case for not retaining the role at all.

CASE 63 AVIATION 1972

Eastern Air Lines Flight 401

On 29 December 1972 an Eastern Air Lines L-1011 descended into the Florida Everglades on a night approach to Miami, killing 101 of the 176 aboard. The cause was attention, not mechanics: all three flight-deck crew became absorbed in a landing-gear indicator bulb that had failed to light, and while they worked it, the autopilot's altitude hold was inadvertently disengaged. The aircraft sank slowly; an altitude alert chimed but went unheeded in the cockpit clutter, and no one was monitoring the flight path. The NTSB findings launched decades of research into attention, monitoring, and cockpit-alert design, and the accident is a formative event behind Crew Resource Management. Eastern 401 is the canonical case of a low-priority task crowding out a life-critical one.

THE FULL CASE, IN FIVE BEATS

Background — Three-crew L-1011 on night Miami approach, with no designed division of attention
What Happened — Crew fixated on a burned-out gear bulb while autopilot disengaged and chime went unheeded
The Investigation — NTSB found crew failed to monitor flight path; trivial trigger produced catastrophic outcome
The Capability Gap — No designed division of attention across crew; alert lacked priority weight to cut through
Aftermath & Reform — Formative event behind Crew Resource Management and prioritized cockpit alerting standards

KEY REFERENCES

- NTSB, Aircraft Accident Report: Eastern Air Lines Flight 401, NTSB-AAR-73-14 (1973) — the night approach and full flight deck.
- NTSB-AAR-73-14 (1973) — the landing-gear-bulb fixation, the inadvertent autopilot disengagement, and 101 deaths of 176 aboard.
- NTSB-AAR-73-14 (1973) — the crew's preoccupation with the landing-gear indication that distracted them from maintaining altitude (paraphrased).

LENS APPLICABILITY

LENS uses Eastern 401 in LEN 5 to teach attention as a designable parameter and in LEN 2 to introduce alert prioritization as a capability deliverable. The case anchors the CRM origin story.

CASE 64 AVIATION 1991, 1994

Boeing 737 Rudder Hardovers

Two Boeing 737s fell out of controlled flight when their rudders reversed — moving opposite to the pilots' input: United 585 near Colorado Springs in 1991 (25 killed) and USAir 427 near Pittsburgh in 1994 (132 killed), 157 in all. The NTSB investigation took years because the failure was rare and unrecoverable. It traced the cause to a thermal-shock condition in the rudder's power-control-unit servo valve — cold-soaked hydraulic fluid hitting a hot valve — that could jam the valve and reverse the rudder. The pilots had no procedure to recognize or recover the failure; the manuals and training never anticipated it, and in that state no available control input could save the aircraft. The 737 rudder cases are the book's case for an unrecoverable failure mode hidden inside a working aircraft.

THE FULL CASE, IN FIVE BEATS

Background — 737 rudder PCU servo valve presumed fault-tolerant by certification, hiding a latent failure mode
What Happened — United 585 and USAir 427 rolled and dove unrecoverably; rudders moved opposite to commanded input
The Investigation — NTSB took years to identify thermal-shock servo-valve jam from cold fluid striking hot valve
The Capability Gap — No recoverable cockpit action existed; gap sat in certification and maintenance, not pilots
Aftermath & Reform — Boeing redesigned PCU, retrofitted fleet, and reshaped how rare unrecoverable modes are hunted

KEY REFERENCES

- NTSB, Uncommanded Flight Control Movements (USAir 427 / 737 rudder PCU), NTSB-AAR-99-01 (1999) — the thermal-shock servo-valve finding.
- NTSB, Aircraft Accident Report: United Airlines Flight 585, NTSB-AAR-92-06 (1992) — the initial (undetermined) Colorado Springs investigation, later reopened.
- NTSB-AAR-99-01 (1999) — the servo-valve reversal mechanism and the absence of any recoverable crew action.

LENS APPLICABILITY

LENS uses the 737 rudder cases in LEN 5 as a capability-limits case (when no human action is recoverable, capability engineering must move upstream) and in LEN 7 for certification governance.

CASE 65 TECHNOLOGY 2024

CrowdStrike Falcon Outage

On 19 July 2024 CrowdStrike pushed a content update to its Falcon endpoint sensor that contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to crash, looping the blue screen of death and requiring hands-on recovery of each device. Roughly 8.5 million machines went down at once — across hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record. CrowdStrike's post-incident review found the content update had not been put through the same automated testing or staged rollout as code: the pipeline treated "content" as a lesser category than "code," though the operating system did not. CrowdStrike is the cybersecurity-vendor analog of Knight Capital, six orders of magnitude larger.

THE FULL CASE, IN FIVE BEATS

Background — Falcon sensor runs in Windows kernel; pipeline treated rapid content updates as lighter than code

What Happened — Faulty content config crashed kernel on 8.5 million machines simultaneously; no staged rollout existed

The Investigation — Post-incident review found content lacked code-grade testing; category boundary did not match operational reality

The Capability Gap — For deployment safety content is code; customers had outsourced a missing safety gate to the vendor

Aftermath & Reform — Staged rollouts, kernel-access review, and scrutiny of vendor concentration risk followed the outage

KEY REFERENCES

- CrowdStrike, Falcon Content Update: Preliminary Post-Incident Review (July 2024) — the content-vs-code testing and staged-rollout gap (paraphrased).
- CrowdStrike PIR (2024) — the configuration-file logic error, the kernel crash, and 8.5 million affected Windows machines.
- Microsoft resilient-engineering analyses and Windows kernel-access review (2024).

LENS APPLICABILITY

LENS uses CrowdStrike in LEN 5 as a categories-and-boundaries capability case and in LEN 2 for the vendor-customer trust architecture: customers trusted CrowdStrike's deployment safety; that trust was load-bearing for the operation of the global economy on a single day.

CASE 66 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

On the night of 26 September 1983 the Soviet Oko early-warning system reported five U.S. intercontinental missiles inbound. The duty officer, Lt. Col. Stanislav Petrov, judged it a false alarm — a real first strike would involve hundreds of missiles, not five — and reported it as such rather than up the launch-decision chain. He was right: sunlight glinting off high-altitude clouds had fooled the satellite's infrared sensors. Petrov's judgment is widely credited with averting nuclear war. The case is the book's strongest positive evidence for human-in-the-loop capability: the automation failed in a mode its designers never anticipated, and a person with contextual judgment and the latitude to override it was the recoverability the system had. Keeping humans in some loops is not nostalgia — it is capability engineering.

THE FULL CASE, IN FIVE BEATS

Background — Soviet Oko satellite system fed launch-on-warning posture; duty officer was the single verification checkpoint

What Happened — Oko reported five inbound U.S. ICBMs; Petrov judged it false because a real strike means hundreds

The Investigation — Sunlight on high-altitude clouds at particular geometry fooled infrared sensors; algorithm later modified

The Capability Gap — No automated check could catch unimagined mode; human contextual judgment plus override authority saved it

Aftermath & Reform — Decision credited with averting nuclear war; case preserved in training as argument for human-in-loop

KEY REFERENCES

- D. Hoffman, The Dead Hand: The Cold War Arms Race and Its Dangerous Legacy (2009) — the Oko system and Soviet launch-on-warning posture.
- Accounts of the 26 Sept. 1983 incident (Hoffman; Petrov interviews) — the five-missile alert and Petrov's assessment.
- Investigations of the Oko incident (incl. V. Aksenov, 2004) — sunlight-on-clouds satellite geometry as the false-positive cause.

LENS APPLICABILITY

LENS uses Petrov in LEN 2 as the positive Human-AI Teaming case at the highest possible stakes: the human role is the recoverability of an automation failure that the designers did not anticipate. The case anchors arguments against full-automation of strategic decisions.

CASE 67 TECHNOLOGY 2018

TSB Bank IT Migration

In April 2018 TSB Bank tried to migrate some five million customer accounts off its former parent Lloyds' systems onto a new platform from its current owner, Sabadell, over a single weekend. When services came back online, nearly every component failed: 1.9 million customers were locked out, some saw strangers' accounts, mortgages vanished, payments bounced. Recovery took months and cost over £330 million; the CEO resigned and the regulator fined the bank. The independent review found the platform had been tested under conditions that did not approximate real load, certified ready by a process that did not challenge the certification, and pushed live against technical recommendations that it was not ready. TSB is the financial-sector analog of Healthcare.gov: a migration shipped without adequate testing because schedule pressure overrode the technical signal.

THE FULL CASE, IN FIVE BEATS

Background — TSB needed to migrate five million accounts off Lloyds onto Sabadell platform in one weekend

What Happened — Nearly every component failed at relaunch; 1.9 million customers locked out, £330M cost, CEO resigned

The Investigation — Slaughter and May found unrealistic load testing, unchallenged readiness certification, override of technical advice

The Capability Gap — Technical signal existed but had no authority; governance let executive schedule overrule a known not-ready

Aftermath & Reform — TSB rebuilt testing and migration governance; FCA penalized the override of technical objection

KEY REFERENCES

- Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures.
- Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO's resignation.
- Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification.

LENS APPLICABILITY

LENS uses TSB in LEN 7 as a corporate-governance case and in LEN 8 for the institutional structure of technical-decision authority. Studio projects compare TSB and Healthcare.gov.

CASE 8 EDUCATION AT SCALE 2014

inBloom

inBloom was a $100-million, Gates-funded shared data infrastructure for U.S. student records — and the technology worked: no bug, no breach, no performance failure. What killed it in about fourteen months was everything around the technology. It launched without consent frameworks, community engagement, transparency about data use, or a way for parents to participate in decisions about their children's data. Parent-privacy groups organized opposition state by state, and nine states withdrew. Analysts read inBloom as the failure of technocratic reform — the assumption that technically sound infrastructure generates its own legitimacy. It is the purest governance failure in the dataset, and the book's clearest argument that in education at scale, stakeholder trust and governance are not optional features but load-bearing structure.

THE FULL CASE, IN FIVE BEATS

Background — Gates-funded \$100M shared student-data store; technically sound, built by enterprise engineers on commercial cloud
What Happened — Launched without consent, engagement, or parent voice; nine states withdrew within fourteen months
The Investigation — Data and Society read it as technocratic reform assuming engineering quality confers legitimacy
The Capability Gap — Stakeholder trust treated as add-on rather than precondition; no patch retrofits trust under fire
Aftermath & Reform — Set shared education infrastructure back years; drove a wave of state student-privacy laws

KEY REFERENCES

- M. Bulger, P. McCormick & M. Pitcan, The Legacy of inBloom, Data & Society Research Institute (2017) — inBloom as a failure of technocratic reform.
- Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting.
- Bulger et al. (2017) — governance, not technology, as the cause; "the technology was not the problem."

LENS APPLICABILITY

LENS uses inBloom in LEN 7 as the canonical governance failure and in LEN 1 as a stakeholder-analysis case. The case anchors the LENS threading commitment that equity and accountability are design commitments, not modules. Studio projects (LEN 10) require students to produce a stakeholder-trust deliverable as a precondition for deployment.

CASE 10 GOVERNMENT 2013

Healthcare.gov Launch

When Healthcare.gov launched on 1 October 2013 it collapsed under load it had never been tested for — about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target. The people assembled to build it knew insurance markets and government programs but not technology product launches; key technical roles went unfilled, no office clearly owned the integration (CMS thought the contractor CGI was the lead integrator; CGI did not), and no end-to-end test was run before launch. The fix-it team that rescued the site became the U.S. Digital Service. At root it was a capability mismatch at scale: the organization lacked the human capabilities the system required, and the governance chain surfaced no signal of the gap before launch. It is the rare governance failure that produced lasting institutional reform.

THE FULL CASE, IN FIVE BEATS

Background — Federal ACA marketplace built under fixed political deadline by teams lacking consumer-launch expertise

What Happened — No clear integrator; no end-to-end test; site collapsed serving 27,000 against seven-million target

The Investigation — GAO and HHS-IG found no one understood project status; governance chain surfaced no readiness signal

The Capability Gap — Capability failure wearing a technology costume; missing match of human capability to system requirement

Aftermath & Reform — Rescue team became U.S. Digital Service; rare case producing durable institutional reform

KEY REFERENCES

- U.S. GAO, Healthcare.gov reports (2014–2016) — the launch, the capability gaps, and the absent end-to-end testing.
- HHS Office of Inspector General, Case Study of CMS Management of the Federal Marketplace, OEI-06-14-00350 (2016) — unclear ownership and the CMS/CGI integration confusion.
- HHS OIG (2016) — “no single person had a clear understanding of the project’s status” (quoted).

LENS APPLICABILITY

LENS uses Healthcare.gov in LEN 1 as a problem-framing case (the technical narrative obscures the capability narrative), in LEN 5 to teach capability-requirements analysis for a large government program, and in LEN 8 to discuss the founding of USDS as organizational learning that survived.

CASE 17 INDUSTRIAL 1984

Bhopal

On the night of 2–3 December 1984, about forty tons of methyl isocyanate gas escaped from a Union Carbide pesticide plant in Bhopal, India — the worst industrial disaster in history. Thousands died within hours; estimates of total deaths run to 15,000–20,000, with roughly half a million exposed or injured. Safety systems had been off-line for months, the plant was understaffed, and workers were inadequately trained to recognize or handle the emergency. Investigators traced the catastrophe to operating errors, design flaws, maintenance failures, training deficiencies, and cost-cutting that endangered safety. Bhopal catalyzed the creation of the U.S. Chemical Safety Board and reshaped industrial-safety regulation for decades. It is the largest-magnitude capability-and-governance failure on record.

THE FULL CASE, IN FIVE BEATS

Background — Union Carbide plant stored bulk MIC under cost pressure; understaffed, safety systems off-line, training thin

What Happened — Water entered an MIC tank; non-operational defenses let forty tons vent over the sleeping city

The Investigation — Investigators cited operating errors, design flaws, maintenance failures, training deficiencies, economy measures endangering safety

The Capability Gap — Capability hollowed across training, maintenance, design, staffing, oversight; no layer owned the integrated whole

Aftermath & Reform — Reshaped industrial safety worldwide; catalyzed the U.S. Chemical Safety Board and process-safety regime

KEY REFERENCES

- Union Carbide and government investigation reports (1985) — MIC storage, the disabled safety systems, and plant understaffing.
- Accounts of the 2–3 Dec. 1984 release — the contested toll (thousands of immediate deaths; 15,000–20,000 total estimates; 500,000 exposed). (Figures vary widely across sources; see AUDIT.)
- New York Times investigation (1985) — “operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety” (quoted).

LENS APPLICABILITY

LENS treats Bhopal in LEN 5 as a multi-failure capability analysis case and in LEN 7 as the largest-magnitude governance failure in the dataset. The Failure → Reform arc to the CSB demonstrates the institution-building pattern at industry scale.

CASE 20 INDUSTRIAL 2017

Grenfell Tower

On 14 June 2017 a fire spread up the exterior of Grenfell Tower, a London public-housing block, killing 72 people. It raced because the tower had been wrapped in combustible aluminium-composite cladding during a refurbishment. The public inquiry found the disaster the culmination of decades of failure: cladding firms engaged in "systematic dishonesty," marketing combustible materials as safe; regulators and inspectors missed effectively banned products across sixteen site visits; and the London Fire Brigade, whose "stay put" advice proved fatal, was unprepared for a cladding fire whose risks earlier incidents had already shown. The failure spanned manufacturers, regulators, inspectors, and responders — each contributing a piece, none owning the whole. Grenfell is the book's case for capability failure distributed across many hands.

THE FULL CASE, IN FIVE BEATS

Background — 1970s tower refurbished with combustible aluminium-composite cladding despite expert cautions against high-rise use

What Happened — Kitchen fire climbed the exterior cladding; stay-put advice held residents inside; seventy-two died

The Investigation — Inquiry found systematic dishonesty by cladding firms; sixteen inspections missed effectively banned products

The Capability Gap — Distributed capability failure; fraud, capture, incompetence, lost memory converged with no integrated owner

Aftermath & Reform — Phase 2 report and government response drove building-safety, cladding, and fire-service reforms

KEY REFERENCES

- Grenfell Tower Inquiry, Phase 1 Report (2019) — the fire's spread up the cladding and the failure of "stay put."
- Grenfell Tower Inquiry, Phase 2 Report (2024) — decades of failure and the combustible-cladding decision.
- Phase 2 Report (2024) — cladding firms' "systematic dishonesty" and the inspection failures across sixteen visits.

LENS APPLICABILITY

LENS treats Grenfell in LEN 7 as the canonical multi-actor governance failure case, in LEN 8 to discuss institutional memory of warnings across the LFB and other agencies, and in LEN 10 as a studio prompt for designing the integrated capability-and-governance deliverable that would have caught the cladding decision.

CASE 21 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

Summit Learning, a personalized-learning platform from Summit Public Schools backed by the Chan Zuckerberg Initiative, was offered free to U.S. districts from 2015 and reached roughly 380 schools and 80,000 students by 2018. By 2019 prominent adopters were withdrawing under parent and student pressure — opt-out campaigns in Brooklyn, cancellations in Cheshire, Kennebunk, and McPherson — amid walkouts and complaints about screen time, disengagement, and data privacy. The pedagogy itself (competency-based progression, projects, mentoring) was defensible and often effective; what failed was deployment governance. There was no evaluation framework districts could read before adopting, no parent-facing data agreement, and no exit path independent of the vendor. Summit is a clean test of the book's claim that the governance architecture must be engineered alongside the tool.

THE FULL CASE, IN FIVE BEATS

Background — CZI-backed personalized-learning platform offered free from 2015 on defensible competency-based pedagogy

What Happened — Reached 380 schools, 80,000 students; by 2019 parent revolts and withdrawals over screen time and privacy

The Investigation — Analysts located failure in deployment governance; no evaluation framework, data agreement, or exit path

The Capability Gap — Working pedagogy with no accountability contract collapsed; same pattern as inBloom recurring at scale

Aftermath & Reform — Districts withdrew, CZI revised outreach; episode became standard caution in ed-tech adoption

KEY REFERENCES

- N. Bowles, "Silicon Valley Came to Kansas Schools. That Started a Rebellion," New York Times (2019) — the parent revolt.
- N. Singer, "The Silicon Valley Billionaires Remaking America's Schools," New York Times (2017) — the CZI/Summit rollout.
- B. Herold, Education Week coverage of Summit / CZI implementation (2019) — district adoptions and withdrawals.

LENS APPLICABILITY

LENS uses Summit Learning in LEN 7 as the foundational consent-and-evidence case for educational technology, and in LEN 10 as a studio prompt for the governance artifacts that any educational-technology adoption decision should produce: a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit pathway that does not depend on the vendor's goodwill.

CASE 22 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

Tennessee's Voluntary Pre-K program enrolled some 18,000 four-year-olds a year, and Vanderbilt researchers studied it with a rare large-scale randomized controlled trial — children admitted by lottery against those who applied but didn't enroll. Through kindergarten the pre-K children showed the expected gains in literacy and vocabulary. By third grade the gains had faded, and by sixth grade the pre-K group did somewhat worse than controls on several academic and behavior measures. The unwelcome result was contested, its methods attacked, and the field largely declined to absorb it. The case is in the book not because pre-K is bad — other programs show durable effects — but because the measurement was unusually rigorous and the discipline's capacity to act on an inconvenient finding was tested and largely failed.

THE FULL CASE, IN FIVE BEATS

Background — Tennessee Voluntary Pre-K served 18,000 four-year-olds; oversubscription enabled a rare lottery-based RCT

What Happened — Kindergarten gains faded by third grade; sixth-grade pre-K children did somewhat worse than controls

The Investigation — Field contested the methods and largely declined to internalize findings, defending policy rather than interrogating

The Capability Gap — Measurement instrument worked; institutional architecture for absorbing inconvenient evidence did not exist

Aftermath & Reform — Episode became methodological touchstone in early-childhood research, argued over more than acted upon

KEY REFERENCES

- M. Lipsey, D. Farran & K. Durkin, "Effects of the Tennessee Pre-Kindergarten Program... Through Third Grade," Early Childhood Research Quarterly 45: 155–176 (2018) — the RCT and fade-out.
- K. Durkin, M. Lipsey, D. Farran & E. Wiesen, "Effects of a Statewide Pre-Kindergarten Program... Through Sixth Grade," Developmental Psychology 58(3): 470–484 (2022) — the sixth-grade reversal (quoted).
- Responses and counter-analyses to the TN-VPK findings (2018–2022) — the contested reception.

LENS APPLICABILITY

LENS uses Tennessee Pre-K in LEN 4 as a measurement-architecture case (longitudinal RCT in a real policy context), in LEN 7 to discuss the institutional politics of unwelcome findings, and in LEN 10 as a studio prompt for designing the implementation-science pathway that would absorb such findings into program redesign rather than rejection.

CASE 35 EDUCATION AT SCALE 2020

UK A-Level Algorithm / Ofqual

When COVID-19 cancelled the UK's 2020 exams, the regulator Ofqual used an algorithm to "standardize" teacher-predicted A-level grades against each school's historical performance. About 39% of grades were adjusted downward and roughly 280,000 entries were downgraded; high-achieving students at historically low-performing state schools were systematically capped by their school's past results, while small-cohort private-school students kept their predicted grades. The algorithm encoded existing structural inequality and amplified it nationwide in a single day. After public outcry — Hannah Fry called it the first time an entire nation felt an algorithm's injustice at once — the government withdrew the algorithm within days and reverted to teacher assessment. It is the book's clearest case of an algorithm deployed without the governance to catch its discriminatory effect.

THE FULL CASE, IN FIVE BEATS

Background — COVID cancelled exams; Ofqual built an algorithm standardizing teacher grades against school historical performance

What Happened — About 280,000 entries downgraded; state-school high achievers capped; private small cohorts kept predictions

The Investigation — Analysts found the model encoded structural inequality; Fry called it a nationally felt algorithmic injustice

The Capability Gap — Missing capability was governance; no equity assessment, sign-off on fairness tradeoff, or appeal path

Aftermath & Reform — Government reversed within days, accepting teacher grades; case became algorithmic-accountability landmark

KEY REFERENCES

- MIT Technology Review, "The UK Exam Algorithm Fiasco" (2020) — Ofqual's standardization approach.
- University of Bristol Centre for Multilevel Modelling and Ofqual analyses — 39% downgraded, 280,000 entries, and the school-prior cap.
- Oxford Internet Institute commentary (2020) — the algorithm as encoded structural inequality ("magical thinking").

LENS APPLICABILITY

LENS uses A-Level in LEN 7 as the foundational case for bias, risk, and governance, in LEN 4 as a measurement-instrument case (an instrument that encodes the bias of its training data), and in LEN 10 as a technical case for bias detection and mitigation. The case is paired in this book with Robodebt (Case 36) and educational algorithmic bias (Case 37).

CASE 36 GOVERNMENT 2016 – 2020

Australia Robodebt

From 2016 the Australian government automated welfare-debt recovery with an income-averaging algorithm that compared welfare records against annual tax data spread evenly across the year. That assumption — steady, year-round employment — fit only about 7% of recipients; the rest, with irregular work, were misclassified as overpaid. Roughly a million debt notices went out, about 470,000 of them later found wholly or partly unlawful, with the burden of proof reversed onto recipients and no human review. The scheme was linked to deaths, including suicides. A 2023 Royal Commission found it sustained by "venality, incompetence and cowardice," and a class action settled for A$1.8 billion. Robodebt is the canonical case of full automation deployed without a human in the loop and without lawful basis.

THE FULL CASE, IN FIVE BEATS

Background — Australia automated welfare-debt recovery from 2016, replacing case review with income-averaging against tax data

What Happened — Averaging fit only seven percent; roughly a million notices issued with reversed burden of proof

The Investigation — Royal Commission found unlawful scheme sustained by venality, incompetence, cowardice; harms entirely predictable

The Capability Gap — Full automation without human-in-the-loop or lawful basis; reversing burden was a governance choice

Aftermath & Reform — Debts refunded; A\$1.8 billion settlement paid; Robodebt entered political memory as byword for automated harm

KEY REFERENCES

- Royal Commission into the Robodebt Scheme, Final Report (2023) — the scheme and its "venality, incompetence and cowardice."
- Royal Commission (2023) — the income-averaging mechanism, the 7%/93% mismatch, the reversed burden of proof, and deaths associated with the scheme.
- Royal Commission (2023) — the finding of unlawfulness and ministerial failure ("a costly failure of public administration," quoted).

LENS APPLICABILITY

LENS treats Robodebt in LEN 7 as the canonical case for automated decision-making without human oversight and in LEN 2 as a Human-AI Teaming failure at the institutional scale: the human role was designed out of the decision loop entirely. The case anchors the program's argument that automated decision systems require deliverable-grade accountability artifacts before deployment.

CASE 37 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

Most U.S. public colleges now use predictive analytics to flag “at-risk” students for early support. Research finds these models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students — and so misdirect the very support the prediction is meant to trigger. The magnitude depends heavily on how “at-risk” is defined, making this partly a construct-definition problem: the choice of what to predict is itself a capability decision with equity consequences. The models inherit historical patterns of discrimination from their training data, and a “flag” can confirm a biased instructor’s low expectations rather than prompt help. It is the educational analog of the UK A-level case — an algorithm built to do good that can allocate help away from the students who need it most.

THE FULL CASE, IN FIVE BEATS

Background — Most U.S. public colleges now use predictive analytics to flag at-risk students for early advising support
What Happened — Research finds racial calibration bias misclassifying Black and Latinx students; magnitude depends on construct
The Investigation — Researchers traced calibration gaps to training data encoding historical discrimination; deficit framing compounds harm
The Capability Gap — Bias lives in the construct definition of at-risk; capability-engineering decision made implicitly through labels
Aftermath & Reform — Ongoing and quiet across hundreds of dashboards; no single collapse forces a reckoning

KEY REFERENCES

- Surveys of predictive-analytics adoption in U.S. higher education — a large majority of public colleges now use some form.
- K. Bird et al., “Are Algorithms Biased in Education? Exploring Racial Bias in Predicting Community College Student Success,” Journal of Policy Analysis and Management 44 (2025), 379–402 — racial calibration bias, 5× higher at the bottom decile depending on the “at-risk” construct.
- D. Gándara, H. Anahideh, M. Ison & L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open (2024) — bias traced to training data encoding historical discrimination.

LENS APPLICABILITY

LENS treats this case as the positive counterpart to Georgia State (Case 39). LEN 4 examines construct definition as the load-bearing measurement choice. LEN 7 examines the governance architecture that determines whose construct gets adopted. LEN 9 covers the technical bias-mitigation methods.

CASE 68 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

The UK Post Office's Horizon accounting system, built by Fujitsu and rolled out in 1999, generated phantom shortfalls in sub-postmasters' branch ledgers. Rather than accept the software was at fault, the Post Office prosecuted them — around 900 over two decades — for theft and false accounting; people were imprisoned, bankrupted, and driven to suicide, in what is now called the most widespread miscarriage of justice in UK history. Internal documents showed engineers had known about Horizon bugs throughout. Convictions began to be quashed in 2021, and a public inquiry continues. The failure ran through the prosecutor and the courts: each accepted "the computer said so" as authoritative because no actor had the standing or expertise to challenge it. Horizon is the book's case for institutional deference to a flawed algorithm.

THE FULL CASE, IN FIVE BEATS

Background — Sub-postmasters bore personal liability for shortfalls reported by Fujitsu's Horizon accounting system from 1999

What Happened — Phantom shortfalls drove around 900 prosecutions for theft; imprisonment, bankruptcy, and suicides followed

The Investigation — Released documents showed engineers knew Horizon could err; convictions began being quashed in 2021

The Capability Gap — No regulator, prosecutor, or court had standing to interrogate Fujitsu's claim of reliability

Aftermath & Reform — Parliament exonerated convictions and opened compensation schemes; the public inquiry continues its work

KEY REFERENCES

- Post Office Horizon IT Inquiry hearings and exhibits (2020–) — the system, the prosecutions, and the human toll.
- Hamilton & Others v. Post Office Limited (Court of Appeal, 2021) — quashed convictions.
- Internal Fujitsu and Post Office documents released through litigation — engineers' knowledge of Horizon bugs.

LENS APPLICABILITY

LENS uses Horizon in LEN 7 as the canonical example of institutional deference to algorithmic output and in LEN 2 for the most extensive multi-decade automation-bias case in the dataset. Studio projects examine what evidentiary architecture would require **interrogating** automated output before acting on it.

CASE 69 HEALTHCARE TECHNOLOGY 2003 – 2018

Theranos

Theranos claimed a blood-testing device that could run hundreds of tests from a finger-stick drop. It did not work. Internal data showed accuracy far below what the company told investors, partners, and patients; to keep up appearances, Theranos ran most patient samples on conventional commercial analyzers and reported them as its own device's results. It reached a $9-billion valuation and put unreliable tests in front of real patients through a Walgreens partnership. The fraud exploited a regulatory seam — the FDA had not validated the device, while the CLIA regime governing the lab did not match the company's product claims — and neither the board nor investors had the depth to challenge it; a journalist did. Elizabeth Holmes was convicted of wire fraud in 2022. Theranos is the book's case for fraud exploiting a governance gap between regulatory regimes.

THE FULL CASE, IN FIVE BEATS

Background — Theranos claimed an Edison device running hundreds of tests from one finger-stick, reaching a nine-billion-dollar valuation
What Happened — The device did not work; Theranos secretly ran samples on commercial analyzers and reported them as its own
The Investigation — Carreyrou's reporting and CMS inspectors surfaced the fraud after the FDA-CLIA seam left validation unowned
The Capability Gap — Neither regulator validated novel clinical claims end-to-end; a celebrity board offered prestige instead of scrutiny
Aftermath & Reform — CLIA certificate revoked, the company collapsed, and Holmes was convicted of multiple wire-fraud counts

KEY REFERENCES

- United States v. Elizabeth Holmes (N.D. Cal., 2018–2022) — the indictment and conviction.
- J. Carreyrou, Bad Blood (2018) — the device's failure and the commercial-analyzer substitution.
- CMS inspection reports and the revocation of Theranos's CLIA certificate (2015–2016).

LENS APPLICABILITY

LENS uses Theranos in LEN 7 for the regulatory-seam failure and in LEN 4 for the measurement-validation gap. The case demonstrates that capability engineering at the boundary between regulatory regimes is itself a governance deliverable.

CASE 70 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

To meet aggressive sales quotas, Wells Fargo employees opened roughly 3.5 million unauthorized customer accounts over years. The behavior was widespread and visible to internal risk and compliance functions, but the bank's response was to fire individual "bad apples" while leaving the incentive structure intact — and that structure was the actual cause: sales targets most employees could not meet by ethical means. The 2016 CFPB and OCC actions brought $3 billion in penalties, the CEO resigned, and the Federal Reserve took the unprecedented step of capping the bank's assets. Wells Fargo is the canonical case of an incentive architecture that manufactured the misconduct the institution then prosecuted at the front line, while insisting the misconduct was individual.

THE FULL CASE, IN FIVE BEATS

Background — Wells Fargo drove cross-selling with aggressive branch quotas that pay and jobs depended on hitting

What Happened — Employees opened about 3.5 million unauthorized accounts; the bank fired front-line staff and kept the incentives

The Investigation — CFPB and OCC actions and a 2017 independent-directors report tied misconduct to the sales-target architecture

The Capability Gap — The governance layer that designed the incentives mistook a designed outcome for individual moral failure

Aftermath & Reform — The Federal Reserve capped Wells Fargo's assets; the case anchored teaching on measurement-gaming and incentive design

KEY REFERENCES

- Consumer Financial Protection Bureau, Consent Order against Wells Fargo (2016) — the unauthorized accounts and penalties.
- Office of the Comptroller of the Currency, Consent Order AA-EC-2016-66 (2016) — unsafe or unsound sales practices tied to the incentive-compensation structure (paraphrased).
- Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017) — how the sales-target architecture drove the conduct.

LENS APPLICABILITY

LENS uses Wells Fargo in LEN 4 as a measurement-gaming case and in LEN 7 for the corporate-governance dynamics that allow such incentives to persist for years. Studio projects redesign the incentive architecture.

CASE 71 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

Volkswagen engineered a “defeat device” into its diesel emissions software: code that detected when a car was on a regulatory test bench and switched on full emissions controls only then. On the road the controls were largely disabled, producing nitrogen-oxide emissions up to forty times the legal limit, across roughly 11 million vehicles for years. The deception was uncovered not by regulators but by a small West Virginia university team comparing real-world to lab measurements. Internal documents showed the defeat device was an institutional decision — a deliberate response to a standard VW’s engineers did not believe they could meet within cost — not a rogue act. VW paid more than $33 billion in penalties and remediation, with criminal convictions. Dieselgate is the book’s case for organized evasion of a measurement system.

THE FULL CASE, IN FIVE BEATS

Background — VW promised clean diesel meeting nitrogen-oxide limits its engineers did not believe they could deliver

What Happened — Defeat-device software detected the test bench and enabled controls only there across about eleven million vehicles

The Investigation — A West Virginia team comparing road to lab emissions exposed the cheat; documents showed institutional authorization

The Capability Gap — The regulator's test ran in a regime the vehicle could detect, inviting the gaming it was meant to catch

Aftermath & Reform — VW paid more than thirty-three billion in penalties; the EU introduced real-world driving emissions testing

KEY REFERENCES

- U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances.
- West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab.
- U.S. Department of Justice Plea Agreement with Volkswagen AG (2017) — the institutional decision, $33B+ in penalties, and convictions (quoted).

LENS APPLICABILITY

LENS uses Dieselgate in LEN 4 as the canonical case for measurement-system evasion and in LEN 7 for the corporate- governance dynamics of engineered fraud. The reform pattern parallels Takata (Case 52).

CASE 72 TECHNOLOGY 2014 – 2018

Cambridge Analytica / Facebook

A Cambridge University researcher's personality-quiz app, taken by about 270,000 people, exploited Facebook's then-permissive Graph API to collect not only the quiz-takers' data but their friends' too — roughly 87 million profiles. The dataset was passed to Cambridge Analytica for political micro-targeting. Facebook's permission architecture had been designed and tested for delivering social experiences, not red-teamed against systematic harvesting; its API contract assumed benevolent developer intent. The architecture worked exactly as designed — the design assumption was wrong. The scandal accelerated the EU's GDPR, helped spur the California Consumer Privacy Act, and produced a $5-billion FTC penalty and consent decree. Cambridge Analytica is the book's case for platform-governance failure: a load-bearing assumption about how an interface would be used, never stress-tested against abuse.

THE FULL CASE, IN FIVE BEATS

Background — Facebook's Graph API let apps collect quiz-takers' data and their friends' data without friends consenting
What Happened — A personality-quiz app taken by 270,000 people harvested about 87 million profiles for political micro-targeting
The Investigation — UK ICO and U.S. FTC investigations, prompted by Guardian reporting, found Facebook responsible for what its design permitted
The Capability Gap — The architecture worked as designed; a load-bearing assumption of developer benevolence was never red-teamed
Aftermath & Reform — GDPR and CCPA accelerated; the FTC imposed a five-billion-dollar penalty and a continuing consent decree

KEY REFERENCES

- U.S. FTC, In the Matter of Facebook, Inc., Consent Order (2019) — the $5B penalty; Facebook gave developers "far more user data than was necessary" (quoted).
- UK Information Commissioner's Office, report on Cambridge Analytica / data analytics in political campaigns (2018) — the scope of the harvesting.
- C. Cadwalladr & E. Graham-Harrison, The Guardian investigation (2018) — the disclosure.

LENS APPLICABILITY

LENS uses Cambridge Analytica in LEN 7 as the foundational case for platform governance and data ethics, and in LEN 1 to teach that "design assumption" is a load-bearing capability deliverable.

CASE 73 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

In 2017 attackers exploited a known, two-month-old vulnerability in Apache Struts on an Equifax web portal — a patch had been available, and Equifax's own security team had told IT to apply it; no one did. Over the next 2.5 months the attackers exfiltrated the personal data of 147 million Americans, and Equifax did not disclose the breach until September. A Senate investigation found systematically inadequate patching, an incomplete asset inventory (so the company did not know which systems needed the fix), and an incident-response function treated for years as a cost center. The CEO resigned and Equifax settled for about $700 million. No single failure caused the breach; cumulative inadequacy across routine cybersecurity work did. Equifax is the data-breach analog of the Sago mine disaster (Case 59).

THE FULL CASE, IN FIVE BEATS

Background — Equifax held identity data on most U.S. adults; security had flagged an Apache Struts patch to IT

What Happened — The unapplied patch let attackers exfiltrate 147 million Americans' data over two and a half months

The Investigation — The Senate subcommittee found systematically inadequate patching and no comprehensive IT asset inventory

The Capability Gap — Routine work; patching, inventory, monitoring, response; was each below standard and starved as a cost center

Aftermath & Reform — A 700-million-dollar settlement funded compensation; patching, inventory, and disclosure timelines rose on the agenda

KEY REFERENCES

- U.S. Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019) — the unpatched Apache Struts vulnerability.
- The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017.
- Senate PSI (2019) — "Equifax lacked a comprehensive IT asset inventory" (quoted).

LENS APPLICABILITY

LENS uses Equifax in LEN 5 for cumulative-inadequacy analysis in cybersecurity and in LEN 7 for the institutional dynamics that allow routine work to be chronically underfunded.

CASE 74 INDUSTRIAL 1981

Hyatt Regency Walkway Collapse

During a crowded tea dance in the atrium of the Kansas City Hyatt Regency on 17 July 1981, two suspended walkways collapsed, killing 114 and injuring 216 — then the deadliest structural collapse in U.S. history. The cause was a connection detail changed during construction: the original design hung the walkways from single long rods; the as-built version used a two-rod arrangement that doubled the load on the upper connection. The structural engineer's office approved the change without recalculating the load, and the connection had in fact been overstressed from the start. Missouri revoked the responsible engineers' licenses, and the case became the foundational engineering-ethics example. The capability gap was institutional: nothing required a change to a load-bearing connection to be re-analyzed by the engineer of record.

THE FULL CASE, IN FIVE BEATS

Background — The Kansas City Hyatt's atrium walkways were originally designed to hang from single continuous steel rods

What Happened — A two-rod construction change doubled the upper connection's load; on 17 July 1981 the walkways fell, killing 114

The Investigation — The National Bureau of Standards found the as-built connection had carried roughly twice its intended load

The Capability Gap — Nothing required the engineer of record to re-analyze a load-bearing change before it was built

Aftermath & Reform — Missouri revoked engineers' licenses; shop-drawing review and engineer-of-record responsibility hardened into rule

KEY REFERENCES

- National Bureau of Standards, Investigation of the Kansas City Hyatt Regency Walkways Collapse, NBSIR 82-2465 (1982) — the doubled-load connection (quoted).
- The 17 July 1981 collapse — 114 killed and 216 injured.
- NBSIR 82-2465 (1982) — the as-built connection inadequate even under the building code.

LENS APPLICABILITY

LENS uses Hyatt Regency in LEN 5 for change-control deliverables and in LEN 7 for the engineering-licensure architecture that holds the engineer of record accountable. Studio projects examine the change-control deliverable.

CASE 75 INDUSTRIAL 2018

FIU Pedestrian Bridge Collapse

On 15 March 2018 a pedestrian bridge under construction at Florida International University collapsed onto the road below, killing six. The NTSB found the design firm had made calculation errors — underestimating demand and overestimating capacity at a critical truss connection — that the independent peer review failed to catch. In the days before, large cracks appeared at that connection; they were inspected, discussed in a meeting the morning of the collapse, and judged not a safety concern, even as crews worked on the span and traffic ran beneath it. The errors were the timeless ones — a connection underdesigned, a review that missed it, warning signs rationalized away — in a structure built with state-of-the-art tools. FIU is the contemporary proof that better design software does not retire the old failure modes.

THE FULL CASE, IN FIVE BEATS

Background — The FIU-Sweetwater pedestrian bridge was a modern, computationally designed concrete truss over a Miami road

What Happened — On 15 March 2018 it collapsed during construction, killing six; NTSB found truss-connection calculation errors

The Investigation — Large cracks were inspected, judged not safety-critical, and reaffirmed in a meeting the morning of collapse

The Capability Gap — More capable design tools did not retire timeless failure modes; review and stop-work rules lagged the software

Aftermath & Reform — NTSB faulted designer, peer reviewer, and parties keeping the road open; the case paired with Hyatt Regency

KEY REFERENCES

- NTSB, Pedestrian Bridge Collapse over SW 8th Street, Miami, Highway Accident Report HAR-19/02 (2019) — the FIGG design calculation errors (quoted).
- NTSB HAR-19/02 (2019) — the 15 March 2018 collapse during construction; six killed.
- NTSB HAR-19/02 (2019) — the observed cracking judged not safety-critical and the morning-of meeting.

LENS APPLICABILITY

LENS uses FIU in LEN 5 to teach review-process deliverables and in LEN 8 to compare with the Hyatt Regency (Case 74) case from forty years earlier — same failure mode, different decade.

CASE 76 ENERGY 2018

Camp Fire / PG&E

On 8 November 2018 a worn hook on a nearly century-old PG&E transmission line failed in high winds and drought, igniting the Camp Fire, which swept into the town of Paradise faster than people could evacuate. Eighty-five died — the deadliest U.S. wildfire in a century — and PG&E later pleaded guilty to 84 counts of involuntary manslaughter. Investigators found PG&E had known for years that its transmission infrastructure across high-fire-risk areas was deteriorating, and had deferred maintenance to fund other priorities, under a regulatory regime that let the deferrals continue. Infrastructure built for one climate was operating in another. Camp Fire is the book's foundational climate-era case for utility capability under changed conditions, and it restructured how California regulates utility wildfire risk.

THE FULL CASE, IN FIVE BEATS

Background — PG&E ran aging transmission lines across drying foothills, with hardware approaching a century old

What Happened — A worn C-hook failed in high winds, igniting a fire that overran Paradise; 85 died, the deadliest in a century

The Investigation — CalFire and the Butte County DA found PG&E had known about deteriorating infrastructure and deferred maintenance

The Capability Gap — Neither utility nor regulator owned the capability to update operations to the actual, hotter, drier risk

Aftermath & Reform — PG&E pleaded guilty to 84 manslaughter counts; California mandated wildfire-mitigation plans and power shutoffs

KEY REFERENCES

- CalFire, Camp Fire Investigation Report (2019) — the worn transmission-line hardware as ignition source.
- The Camp Fire record — 85 killed; PG&E's guilty plea to 84 counts of involuntary manslaughter (2020).
- Butte County District Attorney, The Camp Fire Public Report (2020) — PG&E's knowledge and deferred maintenance (quoted).

LENS APPLICABILITY

LENS uses the Camp Fire in LEN 7 for utility regulatory governance and in LEN 8 for institutional response to changing operating conditions. The case is paired with the Northeast Blackout (Case 62) as utility-capability failures of different kinds.

CASE 77 ENERGY 2021

Texas Grid Freeze

In February 2021 Winter Storm Uri drove temperatures across Texas far below what the ERCOT grid's generators and gas supply were built to withstand. Gas wells froze, gas-fired generation (the bulk of the lost capacity) tripped, and coal, nuclear, and wind also fell; over roughly four days ERCOT shed load and millions lost power in blackouts that, in many cases, never rolled. Texas officially recorded 246 deaths; peer-reviewed excess-mortality estimates run 700–1,000, with about $130 billion in damage. The proximate failure was the absence of winterization requirements for Texas generators — and ERCOT had been warned, after similar storms in 1989, 2011, and 2014, that winterization was inadequate. The warnings never produced a mandate, because the cost would pass to consumers and the rare event was judged acceptable.

THE FULL CASE, IN FIVE BEATS

Background — Texas ran its ERCOT grid outside federal reliability jurisdiction with winterization left optional for operators
What Happened — Winter Storm Uri froze gas wells and tripped generation; about 246 died officially and damage reached 130 billion
The Investigation — The FERC-NERC joint inquiry named absent winterization as the proximate failure, with warnings from 1989, 2011, 2014
The Capability Gap — Optional winterization persisted because costs would pass to consumers and the rare event was judged acceptable
Aftermath & Reform — Texas mandated weatherization of generation and gas with inspections and penalties behind the new requirements

KEY REFERENCES

- FERC–NERC, The February 2021 Cold Weather Outages in Texas and the South Central United States (Joint Inquiry, 2021) — winterization as the proximate failure (quoted).
- Texas DSHS official death count (246); peer-reviewed excess-mortality estimates (700–1,000); $130 billion in damage.
- FERC–NERC (2021) — the prior cold-weather warnings (1989, 2011, 2014).

LENS APPLICABILITY

LENS uses the Texas Grid Freeze in LEN 7 as a regulatory-governance case and in LEN 8 for institutional learning across repeated near- misses (1989, 2011, 2014) that did not produce remediation.

CASE 32 GOVERNMENT 2014

VA Wait-Time Scandal

In 2014 the VA Inspector General found that staff at the Phoenix VA — and then nationwide — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care; some died waiting while the system reported success. A 14-day access target, unrealistic given staffing, was met by hiding reality rather than surfacing it. The warning signs ran back fifteen years: GAO had flagged data-reliability problems since 2000, the IG since 2005, with no systemic fix — and schedulers, the staff operating the measurement, are among the VA's highest-turnover roles, so the institution kept losing the knowledge even to see it was lying to itself. The VA case is the book's canonical example of normalization of deviance applied to measurement itself.

THE FULL CASE, IN FIVE BEATS

Background — A 14-day access target pressed hardest on schedulers, the highest-turnover front-line role
What Happened — Phoenix and nationwide staff hid waits on secret lists while official metrics reported success
The Investigation — Fifteen years of GAO and IG warnings landed on a continually relearning scheduler workforce
The Capability Gap — Measurement gaming hardened into routine practice once turnover erased memory of deviance
Aftermath & Reform — Resignations and the 2014 Choice Act followed; trustworthy measurement proved harder to rebuild

KEY REFERENCES

- VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data.
- GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings.
- VA OIG reports (2005, 2007, 2008) — prior, unactioned findings.

LENS APPLICABILITY

LENS treats the VA case in LEN 4 as the canonical evidence-gap case (the measurement system itself was the source of harm), in LEN 7 as a governance failure (multiple warnings unactioned over fifteen years), and in LEN 8 as a knowledge-loss case via turnover.

CASE 38 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

The Generalized Intelligent Framework for Tutoring (GIFT) is an open-source framework, originated at the U.S. Army Research Laboratory, for authoring and delivering intelligent tutoring systems. Computer-based tutoring has been shown to be about as effective as expert human tutors in well-defined domains, and GIFT exists to lower the barrier to building it; the framework is actively developed, with regular releases and a peer-reviewed annual symposium. The puzzle is not that GIFT failed — it did not — but that ubiquitous fielded adoption across routine military training remains limited despite decades of supporting research. That gap is the canonical learning-engineering problem: the science is settled, the platform exists, the studies are positive — and the institutional pathway to scaled adoption is still being built.

THE FULL CASE, IN FIVE BEATS

The Shift — Decades of research settled efficacy; the open question became why adaptive tutoring isn't everywhere

What Is Emerging — ARL and DEVCOM sustain GIFT as open-source authoring infrastructure with releases and symposium

The Capability Question — Ubiquitous fielded adoption across routine military training remains limited despite the working framework

Early Evidence — Tutoring-effectiveness literature is robust and GIFT-based studies continue to show learning gains

Open Problems — Procurement, pipeline integration, instructor workflows, and authority structure remain unbuilt adoption artifacts

KEY REFERENCES

- K. VanLehn, “The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems,” Educational Psychologist 46(4) (2011) — tutoring effectiveness.
- GIFT Project, gifttutoring.org — the framework, releases, and development under ARL / DEVCOM.
- Editors’ synthesis of the GIFT adoption record — active development but limited ubiquitous fielding (quoted).

LENS APPLICABILITY

LENS treats GIFT in LEN 1 as the problem-framing case for the discipline, in LEN 8 as the foundational adoption-pathway case, and in LEN 10 (capstone) as a prompt for designing the institutional deliverables that would close an adoption gap of this shape.

CASE 40 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

The Experience API (xAPI) was created to track learning experiences across platforms, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competencies, and credentials flowing across organizational boundaries to support continuous capability development — the kind of evidence infrastructure LENS depends on. In practice, xAPI implementations remain largely siloed inside individual learning-management systems; the cross-organizational data sharing that matters most for high-consequence domains has not materialized at scale. The technical standard exists and reference implementations exist; what lags is governance — who owns the data, what consent applies, how quality is assured. xAPI mirrors inBloom (Case 8) in structure — technology ahead of governance — across the whole learning-technology ecosystem. It is the book's case for treating data governance as a capability deliverable.

THE FULL CASE, IN FIVE BEATS

The Shift — Learning crosses many systems, so the field needed common records of experiences
What Is Emerging — xAPI and ADL's Total Learning Architecture envisioned records and credentials flowing across organizations
The Capability Question — Implementations remain siloed inside individual LMS platforms; cross-organizational data sharing has not scaled
Early Evidence — Data model and reference implementations work; ownership, consent, and cross-organization quality assurance lag
Open Problems — Like inBloom, a sound standard arrived ahead of the governance needed for sharing

KEY REFERENCES

- Advanced Distributed Learning Initiative, Total Learning Architecture documentation — the cross-boundary vision.
- xAPI specification (xapi.com) — the technical standard and reference implementations.
- IEEE ICICLE proceedings on TLA adoption challenges — the persistence of siloed implementations.

LENS APPLICABILITY

LENS treats xAPI/TLA in LEN 4 as a data-governance and interoperability case and in LEN 8 as an example of organizational- learning infrastructure that has not scaled. The case is the technical underlay to the larger argument about evidence systems that decision-makers can trust.

CASE 41 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

Implementation science has a canonical finding: it takes an average of about seventeen years for research evidence to reach clinical practice, and only roughly 14% of research findings ever make it at all. This is not a single incident but a systemic condition — effective interventions exist; the system to adopt, sustain, adapt, and measure them at scale does not. Frameworks like the Active Implementation Frameworks (Fixsen et al., 2005) and EPIS (Aarons et al., 2011) were built specifically to attack this gap, and LENS threads implementation science throughout its curriculum in direct response. The seventeen-year gap is the meta-case for the whole book: every success case is a closure of this gap in one domain, every failure case the gap left open. The discipline exists to make seventeen years shorter.

THE FULL CASE, IN FIVE BEATS

The Shift — Implementation science arose because knowing what works and practicing it are different problems
What Is Emerging — Translation averages seventeen years, and only about fourteen percent of findings ever reach practice
The Capability Question — Effective interventions exist; the system to adopt, sustain, adapt, and measure them does not
Early Evidence — Active Implementation Frameworks and EPIS show implementation can be engineered rather than left to chance
Open Problems — Building, funding, and owning the adoption-and-measurement pathway is the general unsolved problem

KEY REFERENCES

- E. A. Balas & S. A. Boren (2000), Yearbook of Medical Informatics — the 17-year / 14% translation figures.
- Z. Morris, S. Wooding & J. Grant, “The answer is 17 years, what is the question,” J. Royal Society of Medicine (2011) (quoted).
- D. Fixsen et al., Implementation Research: A Synthesis of the Literature (2005) — the Active Implementation Frameworks.

LENS APPLICABILITY

LENS uses this case in LEN 1 as the foundational problem statement of the discipline, in LEN 10 as a studio constraint (designs must consider implementation, not just efficacy), and in LEN 8 as the central knowledge-transfer challenge. The case is referenced at least once in every required course.

CASE 78 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

When NASA returned to heavy-lift rockets in the 2000s, it confronted an uncomfortable fact: the institutional capability to build a Saturn V — the rocket that sent Apollo to the Moon — had been lost. The drawings and documents survived; the practical knowledge of how the components were manufactured, tested, and assembled, and why each choice was made, had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned but not reproduced. Apollo was documented to an unprecedented degree, yet the documentation was not the capability: when the engineers who built the system left, the system left with them. Saturn V is the book's strongest evidence that institutional capability lives in people and sustaining practices, not in the artifacts an institution leaves behind.

THE FULL CASE, IN FIVE BEATS

Background — Saturn V was one of history's best-documented engineering programs with exhaustive drawings and records
What Happened — By the 2000s NASA could redesign Saturn V but no longer reproduce its making-knowledge
The Investigation — Documentation captured the what; the tacit how walked out with the retired workforce
The Capability Gap — Institutional capability lives in people and practices, not in the artifacts they leave behind
Aftermath & Reform — Serious programs now use apprenticeship and team continuity to keep tacit capability alive

KEY REFERENCES

- NASA Constellation Program documentation and reviews (2005–2010) — the difficulty of reproducing Saturn V capability.
- R. E. Bilstein, Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles (NASA SP-4206, 1980) — the program and its workforce.
- The documentation-vs-capability distinction (editors' synthesis of the Saturn V record).

LENS APPLICABILITY

LENS uses the Saturn V case in LEN 8 as the foundational organizational-memory case and in LEN 1 to teach the distinction between recorded knowledge and institutional capability.

CASE 79 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

Boeing's Starliner, meant to be the second U.S. commercial crew vehicle alongside SpaceX's Crew Dragon, accumulated failures across a decade: software errors on its 2019 uncrewed flight, valve corrosion scrubbing a 2021 launch, and propulsion-system trouble on its 2024 crewed test that left two NASA astronauts on the ISS rather than returning them as contracted — NASA brought them home on SpaceX months later. GAO and NASA reviews found capability erosion across multiple engineering disciplines at a contractor whose human-spaceflight record had once been definitive. The erosion looks partly generational (Apollo- and Shuttle-era engineers retired) and partly institutional (cost and schedule pressure, and the thinning of NASA's in-house depth to challenge the contractor). Starliner is the contemporary case for capability erosion at a legacy contractor.

THE FULL CASE, IN FIVE BEATS

Background — Boeing won Commercial Crew on a human-spaceflight heritage once definitive in American aerospace
What Happened — Software errors in 2019, valve corrosion in 2021, and 2024 propulsion trouble stranded astronauts
The Investigation — GAO and NASA found generational retirement plus cost pressure eroding both contractor and buyer
The Capability Gap — Reputation outran reality because no instrument measured the legacy contractor's current capability
Aftermath & Reform — NASA leaned on independent reviews and SpaceX as the verified alternative absorbing the failure

KEY REFERENCES

- U.S. GAO, NASA Commercial Crew Program, GAO-20-121 (Jan. 2020) and GAO-19-504 (2019) — schedule slips and technical risks across the program.
- The Starliner test history — 2019 software errors, 2021 valve scrub, and the 2024 crewed flight and ISS stay.
- NASA reviews and reporting on the Starliner test program — issues across software, valves, propulsion, and integration testing (editors' synthesis).

LENS APPLICABILITY

LENS uses Starliner in LEN 8 as a contemporary contractor- capability erosion case and in LEN 5 for the contractor-NASA interface capability that has thinned over decades.

CASE 80 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

On its 1996 maiden flight, the Ariane 5 rocket destroyed itself 37 seconds after launch, losing around half a billion dollars in payloads. The cause was reused software: the inertial reference system inherited code from Ariane 4, where a horizontal-velocity value never exceeded a 16-bit integer's range. Ariane 5′s steeper, faster trajectory pushed it past that range; the integer overflow crashed both redundant reference systems almost simultaneously, the vehicle lost guidance and broke up. The inquiry found the offending code path had been irrelevant on Ariane 4 and was never removed or re-verified for Ariane 5 — software reuse had been treated as risk-free. Ariane 5 is the foundational safety-critical- software case for the hazard of reusing code without re-verifying it against the new system's operating envelope.

THE FULL CASE, IN FIVE BEATS

Background — Ariane 5 reused inertial reference software flown reliably on Ariane 4 to reduce risk
What Happened — The 1996 maiden flight veered off course and broke up 37 seconds after launch
The Investigation — A horizontal-velocity overflow shut down both redundant reference systems simultaneously through identical inherited code
The Capability Gap — Code is fit only for the envelope it was verified against; reuse demands re-verification
Aftermath & Reform — Safety-critical reuse became a verification event with every inherited assumption explicitly re-checked

KEY REFERENCES

- J. L. Lions (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996) — the data-conversion overflow (quoted).
- Lions Report (1996) — the 37-second breakup and the lost payloads.
- Lions Report (1996) — the 64-bit-to-16-bit conversion overflow and the simultaneous shutdown of both inertial reference systems.

LENS APPLICABILITY

LENS uses Ariane 5 in LEN 5 for software-engineering capability deliverables and in LEN 8 for the institutional discipline that treats reuse as a verification event rather than as a savings.

CASE 81 INDUSTRIAL 1940

Tacoma Narrows Bridge

The Tacoma Narrows Bridge over Puget Sound twisted itself apart on 7 November 1940, four months after opening, in a wind no stronger than a stiff breeze and well within its design speed. The piers and cables were sound; the failure mode was aeroelastic flutter — a wind-driven oscillation that the bridge's slender, shallow deck (chosen for elegance and economy over a deeper, stiffer truss) was prone to, and that the discipline had not yet learned to recognize in long-span suspension bridges. No one died, and the collapse was captured on film shown to nearly every civil-engineering student since. The cost of the lesson was a bridge; the payoff is that no comparable bridge has failed the same way. Tacoma Narrows is the book's clearest example of an entire discipline learning permanently from one failure.

THE FULL CASE, IN FIVE BEATS

Background — The 1940 bridge used a slender shallow deck chosen for elegance and economy
What Happened — A 40 mph wind twisted the deck apart while piers and cables held
The Investigation — Aeroelastic flutter was a self-amplifying oscillation static load analysis could not anticipate
The Capability Gap — The missing capability lived above any engineer in a discipline not yet knowing flutter
Aftermath & Reform — Wind-tunnel testing, aeroelasticity teaching, and the collapse film locked the lesson into the profession

KEY REFERENCES

- O. Ammann, T. von Kármán & G. Woodruff, The Failure of the Tacoma Narrows Bridge (1941) — the failure inquiry.
- The 7 Nov. 1940 collapse in a 40 mph wind and the slender-deck design choice.
- K. Y. Billah & R. Scanlan, “Resonance, Tacoma Narrows Bridge Failure, and Undergraduate Physics Textbooks,” American Journal of Physics (1991) — the aeroelastic-flutter analysis.

LENS APPLICABILITY

LENS uses Tacoma Narrows in LEN 1 as a problem-framing exemplar for discipline-level learning, in LEN 8 as the canonical case for successful institutional knowledge absorption, and in studio (LEN 10) as a worked example of what discipline-level reform requires.

CASE 82 ENERGY 2015 – 2016

Aliso Canyon Methane Leak

In October 2015 a well at SoCalGas's Aliso Canyon underground gas-storage field ruptured and could not be capped for nearly four months, releasing roughly 100,000 tons of methane — the largest such leak in U.S. history, equivalent to the annual emissions of half a million cars — and forcing the evacuation of about 8,000 households in Porter Ranch. The root-cause analysis found external corrosion of the well casing, a subsurface safety valve removed in the 1970s and never replaced, and a pattern of deferring integrity checks on the field's aging wells. The operator knew the wells were old and their integrity uncertain; so did the regulator; the inspection regime had not been engineered to catch the failure mode that occurred. Aliso Canyon is the book's case for legacy infrastructure aging past its oversight.

THE FULL CASE, IN FIVE BEATS

Background — SoCalGas stored gas in 1950s oil wells whose subsurface safety valve was long removed

What Happened — Well SS-25 ruptured in October 2015, venting 100,000 tons and evacuating thousands of households

The Investigation — External corrosion was a known inspectable mode the operator had deferred on aging wells

The Capability Gap — Operator and regulator both knew the wells were aging; no rule forced inspection

Aftermath & Reform — California tightened rules with required inspections, pressure limits, and risk-management plans

KEY REFERENCES

- Blade Energy Partners, Aliso Canyon Root Cause Analysis (2019) — external corrosion of the well casing (quoted).
- The leak record — 100,000 tons of methane; 8,000 households evacuated; the largest U.S. methane leak.
- Blade Energy (2019) — the removed subsurface safety valve and the deferred integrity checks on aging wells.

LENS APPLICABILITY

LENS uses Aliso Canyon in LEN 7 for regulatory-architecture capability and in LEN 8 for the institutional pattern of legacy assets aging past the regime governing them.

CASE 31 HEALTHCARE ONGOING

Medical Errors as Systemic Failure

Medical error in the United States is not a single incident but a systemic condition that the system's own measurement instruments cannot see. In 2016, Makary and Daniel of Johns Hopkins estimated, in the BMJ, that medical errors cause more than 250,000 deaths a year — which would make them the third leading cause of death behind heart disease and cancer. Death certificates do not record medical error as a cause, so the problem is structurally invisible to the systems meant to track it. The 250,000 figure has been substantively contested on methodological grounds — a dispute that is itself a worked example of the gap-attribution problem. The Institute of Medicine raised the alarm in 1999; seventeen years later the problem persisted at scale.

THE FULL CASE, IN FIVE BEATS

Background — Death certificates record proximate physiology with no field for medical error
What Happened — Makary and Daniel estimated 250,000 annual U.S. error deaths from systemic care failings
The Investigation — Critics challenged the extrapolation, arguing counterfactual attribution of error-caused death resists clean tallying
The Capability Gap — Without a tracked instrument and robust attribution, safety programs cannot prove worth against invisible harm
Aftermath & Reform — Targeted reforms cut bounded harms while system-wide mortality kept escaping measurement and persisted at scale

KEY REFERENCES

- Makary, M. & Daniel, M. (2016), "Medical error — the third leading cause of death in the US," BMJ — the 250,000 estimate and the quoted framing.
- Makary & Daniel (2016), BMJ — death certificates do not capture medical error as a cause.
- Shojania, K. & Dixon-Woods, M. (2017), "Estimating deaths due to medical error," BMJ Quality & Safety — methodological contestation.

LENS APPLICABILITY

LENS treats this case as the central evidence anchor of the curriculum: LEN 1 as the foundational problem statement, LEN 4 as the canonical case for measurement systems blind to their own failures (death certificates do not capture the failure mode), and LEN 10 as the studio prompt for interventions at population scale.

CASE 83 GOVERNMENT TECHNOLOGY 2003 – 2012

LIBOR Manipulation

LIBOR — the London Interbank Offered Rate — was the benchmark interest rate referenced in roughly $350 trillion of financial contracts, yet it was set each day not from observed transactions but from a small panel of banks’ estimates of what they would pay to borrow. For about a decade, banks systematically shaded those estimates to favor their own derivatives positions; the coordination was later laid bare in subpoenaed trader-and-submitter messages. The capability gap was in the design of the benchmark itself: it asked for declarations, not observations, and the architecture invited gaming. Regulators levied more than $9 billion in fines, and LIBOR was eventually replaced by SOFR, a rate built from actual transactions. The reform changed the unit of measurement, not just the rules.

THE FULL CASE, IN FIVE BEATS

Background — Benchmark set from panel banks' rate estimates rather than observed transactions
What Happened — Traders pressured submitters to shade rates favoring their derivatives books for years
The Investigation — DOJ and UK regulators levied over nine billion in fines; Wheatley Review blamed design
The Capability Gap — Declarations rather than observations made systematic gaming the architecture's predictable output
Aftermath & Reform — LIBOR phased out; SOFR built on observed Treasury repo transactions replaced it

KEY REFERENCES

- U.S. Department of Justice settlements with Barclays, UBS, Deutsche Bank, RBS et al. (2012–2015) — the manipulation and fines.
- Wheatley, M., The Wheatley Review of LIBOR: Final Report (2012) — the estimate-vs-transaction design flaw (paraphrased).
- Wheatley Review (2012) — recommendation to anchor benchmarks in observed transactions.

LENS APPLICABILITY

LENS uses LIBOR in LEN 4 as the canonical measurement-architecture case at financial-system scale and in LEN 7 for the cross- jurisdictional governance reform that produced SOFR. The case pairs with Wells Fargo (Case 70) and Volkswagen (Case 71) as measurement-gaming cases.

CASE 84 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

Under a celebrated superintendent, Atlanta Public Schools was held up nationally as a high-performing urban district. The performance was substantially fabricated. A state special investigation, supported by the Georgia Bureau of Investigation, and reporting by the Atlanta Journal-Constitution found that for several years educators across dozens of schools had systematically erased and corrected students' answers on state standardized tests. The cheating was organized — principals pressured teachers, staff held "erasure parties" after testing days — and the incentive system rewarded the gaming with bonuses and promotions. Around 180 educators were implicated, 35 indicted, and 11 convicted under Georgia's racketeering statute. The capability gap lay in the measurement architecture: the institution being measured operated the instrument that measured it, with no independent audit.

THE FULL CASE, IN FIVE BEATS

Background — District administered and reported the very high-stakes tests determining bonuses and recognition
What Happened — Educators across dozens of schools organized erasure parties to change students' answers
The Investigation — Newspaper analysis and state investigators documented scheme; thirty-five indicted, eleven convicted under RICO
The Capability Gap — Institution being measured operated the instrument measuring it, with no independent audit
Aftermath & Reform — Convictions made APS the prominent U.S. high-stakes-testing fraud case fuelling reassessment

KEY REFERENCES

- Special Investigators, Final Report on Atlanta Public Schools (2011) — the organized cheating and the quoted finding.
- Atlanta Journal-Constitution investigative series (2009–2011) — the erasure-rate analysis.
- State of Georgia v. Hall et al. (2014–2015) — indictments and RICO convictions.

LENS APPLICABILITY

LENS uses APS in LEN 4 as the foundational U.S. education measurement-gaming case and in LEN 7 for the incentive- architecture failure. Studio projects examine how an independent audit would have caught the cheating.

CASE 85 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

Bernard Madoff ran the largest Ponzi scheme in history — roughly $65 billion in fictitious account value — for years while the SEC repeatedly investigated and cleared him. Financial analyst Harry Markopolos delivered the agency a detailed memo in 2005, "The World's Largest Hedge Fund is a Fraud," showing that Madoff's steady returns were mathematically impossible. The SEC opened an investigation and concluded no action was warranted. Madoff operated until his sons turned him in during the 2008 crisis, when redemptions became impossible to honor. The SEC Inspector General later found the staff assigned lacked the expertise to evaluate Markopolos's arguments and had deferred to Madoff's industry stature. The capability gap was at the regulator's technical-evaluation pipeline, not at the evidence — which was specific and checkable.

THE FULL CASE, IN FIVE BEATS

Background — Former NASDAQ chairman reported steady returns that were entirely fictitious Ponzi fabrications

What Happened — Markopolos delivered SEC a detailed quantitative memo in 2005; investigation closed without action

The Investigation — SEC Inspector General found numerous missed opportunities and staff deference to Madoff's stature

The Capability Gap — Regulator's technical-evaluation pipeline lacked people able to check the checkable math

Aftermath & Reform — Collapse prompted SEC reforms including Office of Market Intelligence for technical triage

KEY REFERENCES

- SEC Office of Inspector General, Investigation of Failure of the SEC to Uncover Madoff's Ponzi Scheme, Report OIG-509 (2009) — the quoted finding.
- Markopolos, H. (2010), No One Would Listen — the 2005 memo and its dismissal.
- United States v. Madoff (2009) — guilty plea and the $65B figure.

LENS APPLICABILITY

LENS uses Madoff in LEN 4 for technical-evidence evaluation and in LEN 7 for regulator-capability deliverables. The case pairs with Theranos (Case 69) as cases where regulators lacked the technical depth to challenge the evidence presented to them.

CASE 86 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

The September 11, 2001 attacks, which killed 2,977 people, were enabled in part by intelligence the U.S. government already held but never integrated. The CIA knew, from a January 2000 meeting in Kuala Lumpur, that two future hijackers had entered the country; it did not tell the FBI. The FBI separately flagged suspicious flight-training activity in Phoenix and Minneapolis in 2001, but that information was never aggregated. Visa issuance, immigration tracking, and watch-listing were each run by a different agency, and the handoffs between them depended on individual initiative that no institution required. The 9/11 Commission called it a "failure of imagination" — a framing critics say understates the structural gap. The reform that followed built the cross-agency architecture, the ODNI and the NCTC, that had not existed.

THE FULL CASE, IN FIVE BEATS

Background — Multiple agencies held pieces; no institution owned integration as a required architectural function

What Happened — CIA tracked future hijackers from Kuala Lumpur; FBI flagged flight training; none aggregated

The Investigation — 9/11 Commission and Joint Inquiry documented specific sharing failures across FBI, CIA, NSA

The Capability Gap — Architecture was missing; shared watch-lists, mandated handoffs, and a fusion body existed nowhere

Aftermath & Reform — 2004 Intelligence Reform Act created ODNI and NCTC, building integration as institutional deliverable

KEY REFERENCES

- National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004) — the quoted "failure of imagination" and the specific sharing failures.
- 9/11 Commission Report (2004) — the Kuala Lumpur tracking and the Phoenix/Minneapolis flagging.
- Joint Inquiry into Intelligence Community Activities Before and After September 11, 2001 (2002) — cross-agency information-sharing failures.

LENS APPLICABILITY

LENS uses 9/11 in LEN 8 as the foundational case for cross- organizational capability and in LEN 7 for the governance architecture of multi-agency systems. Studio projects compare 9/11 with Eagle Claw (Case 46) as institutional-architecture failures of different kinds.

CASE 87 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

Merck's painkiller Vioxx (rofecoxib) was approved by the FDA in 1999 and prescribed to millions. A 2000 trial, VIGOR, found a roughly five-fold higher rate of heart attacks in the Vioxx arm than in the comparison drug — a signal many read as a protective effect of the comparator rather than a cardiovascular risk of Vioxx. Not until the placebo-controlled APPROVe trial was halted in 2004 was the risk established and the drug withdrawn. For nearly four years, the disclosure architecture between Merck's internal data, FDA reviewers, and prescribers failed to surface the magnitude of the danger to a decision point; an FDA scientist estimated tens of thousands of excess cardiovascular events. The reforms that followed — REMS and the Sentinel Initiative — built post-market surveillance that had not existed.

THE FULL CASE, IN FIVE BEATS

Background — Widely prescribed COX-2 inhibitor approved 1999; post-market surveillance thin relative to exposure

What Happened — VIGOR trial showed five-fold heart attack rate; Merck read comparator as protective instead

The Investigation — Hearings and Graham testimony showed cardiovascular signals present in trial record years earlier

The Capability Gap — Disclosure architecture between manufacturer data, FDA reviewers, and prescribers failed to aggregate signal

Aftermath & Reform — Merck settled near five billion; REMS and Sentinel built active post-market surveillance

KEY REFERENCES

- Graham, D. J. et al. (2005), Lancet — population-level cardiovascular risk analysis.
- Bombardier, C. et al. (2000), VIGOR trial, NEJM — the five-fold myocardial-infarction signal.
- Bresalier, R. et al. (2005), APPROVe trial, NEJM — placebo-controlled confirmation of cardiovascular risk.

LENS APPLICABILITY

LENS uses Vioxx in LEN 4 for post-market surveillance as a measurement deliverable and in LEN 7 for the corporate-regulator dynamics that allowed signal aggregation to be deferred.

CASE 12 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

In March 1977, two 747s collided in fog at Tenerife and 583 people died — in part because a KLM flight engineer's twice-voiced doubt about the runway was overridden by his captain. The failure was not of skill but of the system by which skill in one seat reached another. Crew Resource Management, formalized by United Airlines in 1981, re-engineered the cockpit as a coordinated team: explicit communication protocols, named authority gradients, structured briefings. Twenty years later the Commercial Aviation Safety Team (CAST) added closed-loop analysis of operational data to find and fix hazards before they caused accidents. The pairing — cultural redesign plus continuous evidence — drove an 83% reduction in U.S. commercial-aviation fatality risk between 1998 and 2008, earning the 2008 Collier Trophy.

THE FULL CASE, IN FIVE BEATS

Background — Tenerife showed crashes came from crew coordination breakdowns, not individual lack of flying skill

The Intervention — United formalized CRM in 1981 with protocols, authority gradients, and structured briefings as procedure

How It Worked — Engineered the interaction system so junior challenges had a named, authorized route to the decision

The Evidence — CAST closed-loop hazard work cut fatality risk 83 percent; Collier Trophy in 2008

What Transferred — Design logic of paired cultural change plus measurement loop exported to surgery and AI systems

KEY REFERENCES

- FAA Advisory Circular 120-51E, Crew Resource Management Training — CRM protocols and authority-gradient training.
- Helmreich, Merritt & Wilhelm (1999), "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology.
- Spanish CIAIAC / ALPA reports on the 1977 Tenerife collision — the overridden crew challenge.

LENS APPLICABILITY

LENS treats CRM/CAST as the foundational success case across the curriculum. LEN 1 uses it as the problem-framing exemplar; LEN 4 uses CAST as the model closed-loop evidence system; LEN 2 uses CRM as the template for redesigning human roles in automated environments — the logic now being applied to AI-augmented systems.

CASE 14 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

Peter Pronovost's central-line checklist has five items — wash hands, clean the skin with chlorhexidine, drape the patient, use full barrier precautions, apply a sterile dressing — and not one of them was unknown to the physicians it governed. The question was never what to do, but whether it would be done every time. The Keystone project, launched across 103 Michigan ICUs in 2004, paired the checklist with a cultural change: nurses were not merely permitted but required to stop the procedure if a step was skipped. That authorization was the load-bearing element. Central-line-associated bloodstream infections fell to near zero, an estimated 1,500 lives and $100 million were saved in eighteen months, and the effect was sustained at ten years.

THE FULL CASE, IN FIVE BEATS

Background — Central-line infections persisted because nurses lacked procedural path to intervene across the authority gradient

The Intervention — Pronovost paired a five-item sterile checklist with a required nurse stop authority

How It Worked — Requirement, not permission, plus an impersonal trigger made the stop usable against senior physicians

The Evidence — CLABSI rates fell near zero across 103 ICUs; effect sustained at ten years

What Transferred — Packaged as the AHRQ CUSP toolkit, replicated in forty states and internationally as paired design

KEY REFERENCES

- Pronovost, P. et al. (2006), "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," NEJM 355 — the trial and the near-zero result.
- Pronovost & Vohr (2010), Safe Patients, Smart Hospitals — the checklist-plus-nurse-authority pairing (paraphrased).
- Lipitz-Snyderman, A. et al. (2011), BMJ — sustained effect at follow-up.

LENS APPLICABILITY

LENS uses Keystone in LEN 4 and LEN 10 as the canonical worked example of measurement linked to intervention. Studio projects require students to specify both halves of the pair, name the authority that authorizes the cultural half, and identify the measurement signal that will confirm or refute the effect. The course treats "is the cultural change actually authorized?" as falsifiable, not stated.

CASE 16 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

Three Mile Island did not produce a reactor accident at the next plant over — it produced an institution. Within months of the 1979 Kemeny Commission report, the U.S. commercial nuclear industry founded the Institute of Nuclear Power Operations on a stark premise: an accident at any single plant would threaten every operator's license, and no utility could engineer its safety capability alone. Funded by the utilities it evaluated and operating without statutory authority, INPO set training and certification standards, accredited every plant's programs through the National Academy for Nuclear Training, and ran peer evaluations in which operators from one utility scrutinized another's control rooms and records. The pre-TMI culture of complacency gave way to mandated vigilance. No U.S. commercial reactor has had a significant INES-level event since.

THE FULL CASE, IN FIVE BEATS

Background — Three Mile Island exposed an industry where lessons at one plant never reached others

The Intervention — Utilities founded INPO within months on the premise one accident threatened everyone's license

How It Worked — Honest peer review by working operators gave a non-statutory body its enforcement weight

The Evidence — No significant INES-level event since founding; fleet-wide performance indicators improved broadly across the industry

What Transferred — Shared exposure, regulatory legitimacy, and peer review crossed borders to WANO after Chernobyl

KEY REFERENCES

- Rees, J. (1994), Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island — INPO's design and the "hostages" premise (paraphrased).
- Report of the President's Commission on the Accident at Three Mile Island (Kemeny Commission, 1979) — the pre-TMI culture.
- Nuclear Energy Institute, "Lessons from the 1979 Accident at Three Mile Island"; National Academy for Nuclear Training — accreditation and peer evaluation.

LENS APPLICABILITY

LENS uses INPO in LEN 8 as the canonical example of industry-level learning: students identify the structural conditions in their own domain that would permit (or block) an INPO-equivalent and design the peer-review architecture required. LEN 1 uses the founding moment — nine months after TMI — to discuss the **speed** a credible response to catastrophe demands.

CASE 13 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

In 2008, Atul Gawande's team and the WHO introduced a single-page, nineteen-item surgical checklist applied at three junctures — before anesthesia, before incision, and before the patient leaves the operating room. Piloted across eight hospitals in eight countries, from Toronto to Tanzania, it nearly halved the surgical death rate (1.5% to 0.8%) and cut serious complications by more than a third — results published in the NEJM in 2009. The artifact was the checklist; the intervention was the system of forced pauses it created, three moments when a moving team had to stop and confirm shared state. A later Ontario study found no mortality benefit after a mandated rollout, surfacing the fidelity question: the artifact works only when the institution authorizes its honest use.

THE FULL CASE, IN FIVE BEATS

Background — Surgical harm was widespread because teams in motion rarely paused to verify shared understanding

The Intervention — WHO and Harvard introduced a nineteen-item checklist piloted across eight hospitals in eight countries

How It Worked — Forced pauses at three junctures required teams to halt and speak confirmations aloud together

The Evidence — NEJM study showed death rate halved and complications dropped a third across all sites

What Transferred — Ontario mandated rollout produced no benefit; the pause works only when genuinely authorized

KEY REFERENCES

- Haynes, A. et al. (2009), "A Surgical Safety Checklist to Reduce Morbidity and Mortality," NEJM — the 1.5%→0.8% result and major-complication reduction.
- WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures.
- Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism.

LENS APPLICABILITY

LENS uses the WHO checklist in LEN 10 (capstone) as the canonical end-to-end design exemplar: a problem identified, a minimal artifact prototyped, a multi-site pilot, a measurement regime, and global scale-out. LEN 4 examines the measurement architecture that made the effect provable.

CASE 15 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

After the fatal 2017 Fitzgerald and McCain collisions (Case 1) exposed gutted seamanship training, the U.S. Navy overhauled how it builds surface-warfare competence. It restored the Surface Warfare Officers School from CD-ROM self-study to classroom and simulator instruction, stood up Maritime Skills Training Centers on both coasts, created ten pass-or-fail career assessments — three of them no-go gates — and adopted aviation-style debriefing. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard; three of the first eighteen were immediately sidelined. The structural change is real and the investment substantial. What is missing is the third half: the GAO has noted the Navy still lacks systematic evaluation of whether the reforms work — a live success in structure, with evidence of effect still outstanding.

THE FULL CASE, IN FIVE BEATS

Background — Two fatal 2017 collisions exposed seamanship training degraded to CD-ROM self-study

The Intervention — Navy restored SWOS classroom instruction, tripled training hours, and created ten career assessments

How It Worked — Reform paired simulators and restored curricula with aviation-style debriefing and explicit gate ownership

The Evidence — GAO noted the Navy still lacks systematic evaluation of whether reforms improve readiness

What Transferred — Live in-progress reform shows mature capability engineering must build measurement infrastructure from the start

KEY REFERENCES

- GAO-21-168 (and GAO-20-154), Navy readiness reform — the lack of systematic outcome evaluation (paraphrased).
- Readiness Reform Oversight Council, One-Year Report (2019) — restored training, assessments, and gates.
- Navy and NTSB reports on the Fitzgerald and McCain collisions (2017–2019) — the training-degradation antecedent.

LENS APPLICABILITY

LENS uses Navy SWO reform in LEN 10 (capstone) as a worked exercise in capability intervention at scale, and in LEN 4 as a case where the measurement infrastructure has lagged the intervention. Students design the evidence regime that should accompany the reform.

CASE 23 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

Between 1970 and 1999, Korean Air had one of the worst safety records in commercial aviation — a loss rate roughly seventeen times United's, more than 700 lives lost — and after the 1997 Guam crash of Flight 801 killed 228, the president of South Korea called it "an embarrassment to the nation." The NTSB traced root causes to cockpit authority gradients: junior officers were culturally unable to challenge a captain's errors. In 2000, Korean Air brought in Delta's David Greenberg to rebuild flight operations — mandating English as the cockpit language, adapting CRM for a high-power-distance culture, and bringing in outside consulting from Boeing and Delta. The airline has had no fatal passenger accident since, winning the 2006 Phoenix Award. Cultural legacy is not destiny when it is deliberately redesigned.

THE FULL CASE, IN FIVE BEATS

Background — Korean Air's pre-2000 loss rate was seventeen times United's, rooted in cockpit authority gradients

The Intervention — Korean Air hired David Greenberg from Delta to mandate English, adapt CRM, and modernize

How It Worked — Switching to English stripped out Korean honorifics that had silenced first officers and absorbed warnings

The Evidence — No fatal passenger accident since the reforms; the 2006 Phoenix Award recognized the turnaround

What Transferred — Cultural legacy is not destiny; locate the binding feature and engineer the medium that carries it

KEY REFERENCES

- NTSB, Aircraft Accident Report: Korean Air Flight 801, Guam (2000) — authority gradients as a root cause.
- Air Transport World Phoenix Award documentation (2006) — recognition of the turnaround.
- Gladwell, M. (2008), Outliers — the Korean Air chapter on power distance and the English-language change.

LENS APPLICABILITY

LENS uses Korean Air in LEN 8 as the canonical organizational- learning case under cultural constraint and in LEN 2 as a CRM- extension case for high-power-distance contexts. The case is paired in this book with the Toyota Andon Cord (Case 24) as cultural intervention success stories.

CASE 24 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

The andon cord lets any assembly-line worker stop Toyota's entire production line on detecting a defect — handing the lowest-ranking person on the floor the power to halt operations worth millions per hour. The cord is trivially cheap; the authority it confers is the design. The case is decisive for capability engineering because when American automakers copied the cord in the 1980s and 1990s, workers were too afraid to pull it: the tool was present, the empowerment was not. Toyota's system works because it pairs the mechanism with a culture of psychological safety, fast supervisor response, no-blame problem-solving, and the codified "Five Whys" method. When the line stops at Toyota, the team treats it as a learning opportunity. The Andon Cord is the manufacturing twin of the Keystone nurse-authority intervention (Case 14).

THE FULL CASE, IN FIVE BEATS

Background — The cheapest moment to fix a defect is when the seer has least standing to halt
The Intervention — Toyota installed a pull-cord letting any worker signal a problem and stop the line
How It Worked — Mechanism paired with psychological safety, rapid supervisor response, no-blame analysis, and Five Whys
The Evidence — American copies failed because workers feared pulling it; Toyota's protected authority is the variable
What Transferred — Authority interventions and technical artifacts are inseparable; the manufacturing twin of Keystone nurse-stop

KEY REFERENCES

- Liker, J. (2020), The Toyota Way (2nd ed.) — the andon cord, the cultural pairing, and the American imitation (paraphrased).
- Spear, S. & Bowen, H. (1999), "Decoding the DNA of the Toyota Production System," HBR — the response protocol and embedded learning.
- Shingo, S. (1989), A Study of the Toyota Production System — the technical mechanism.

LENS APPLICABILITY

LENS uses the Andon Cord in LEN 2 as the foundational example of paired technical-cultural intervention, in LEN 8 to discuss cross-domain transfer (and why imitation without the cultural half fails), and in LEN 10 as a studio exemplar of minimal-artifact design.

CASE 27 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

TeamSTEPPS — developed jointly by the Agency for Healthcare Research and Quality and the Department of Defense and released in 2006 — is the healthcare analog of Crew Resource Management (Case 12): an evidence-based team-training framework distilled from fifty years of aviation, military, and nuclear research and adapted for clinical settings. It trains four core competencies: communication, leadership, situation monitoring, and mutual support. It is explicitly the translation pathway from high-reliability research into bedside practice — the cross-domain capability transfer LENS is built to teach. Because its implementation infrastructure was funded as part of the program, TeamSTEPPS moved from research to scaled deployment in years rather than decades, and has been adopted by thousands of healthcare organizations with measurable gains in teamwork and safety culture.

THE FULL CASE, IN FIVE BEATS

Background — High-reliability research showed teamwork drives safety, but clinical settings lacked a structured curriculum

The Intervention — AHRQ and DoD jointly released TeamSTEPPS in 2006, training four core team competencies

How It Worked — Funding master-trainer programs, toolkits, and a support center alongside the curriculum compressed the gap

The Evidence — Studies show improved teamwork and safety culture; on-time surgical first starts rose 21 percent

What Transferred — Cross-domain capability transfer is engineerable when the funded path to sustained practice is included

KEY REFERENCES

- AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies.
- DoD / AHRQ partnership documentation — the joint development and implementation infrastructure.
- Salas, E., Rosen, M. et al. (2009) — cross-domain team-training evidence base.

LENS APPLICABILITY

LENS treats TeamSTEPPS in LEN 8 as the canonical cross-domain transfer case and in LEN 1 as evidence that the program's core proposition — that learning engineering is a discipline — has institutional precedent. Studio projects (LEN 10) reference TeamSTEPPS as the worked example of cross-domain adaptation methodology.

CASE 28 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

Admiral Hyman Rickover built a training and qualification culture for the Naval Nuclear Propulsion Program that has produced zero reactor accidents across more than 60 years and thousands of reactor-years of operation, often in extreme conditions. Every nuclear-trained sailor must pass rigorous qualification to zero-defect standards and demonstrate competence in oral examination by senior nuclear-qualified officers; the culture demands personal accountability, deep technical mastery, and a questioning attitude that obliges operators to challenge assumptions, including superiors'. The sharpest contrast in this book is internal: the same Navy that ran the nuclear program to this standard let surface-warfare training decay to CD-ROMs and paid for it at Fitzgerald and McCain (Case 1). Same institution, two philosophies — and the choice shows up in the casualty columns.

THE FULL CASE, IN FIVE BEATS

Background — Reactors at sea allowed no accident rate, forcing the human operating system to extreme standards
The Intervention — Rickover required every nuclear sailor to pass zero-defect qualification and continuous re-examination
How It Worked — Technical mastery paired with a mandatory questioning posture that obliges challenging superiors' assumptions
The Evidence — Zero reactor accidents across sixty years and thousands of reactor-years; duration is the key evidence
What Transferred — Same Navy let surface training decay; internal contrast isolates training philosophy as the variable

KEY REFERENCES

- Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover's philosophy (paraphrased).
- Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record.
- Admiral Hyman G. Rickover, "Doing a Job" (Columbia University commencement address, 1982) — "people, not organizations... get things done" (quoted).

LENS APPLICABILITY

LENS treats the Rickover model in LEN 8 as the foundational organizational-learning case and in LEN 5 as a worked example of capability requirements traceable from operational analysis through qualification standards. The internal Navy comparison anchors the program's core argument about capability as a system parameter.

CASE 39 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

Georgia State University built a predictive-analytics advising system that tracks some 800 risk factors per student and triggers proactive outreach when early warning signs appear. Crucially, it was designed with equity as a primary constraint — the explicit goal was to eliminate, not reproduce, graduation gaps — and the alerts prompt human advisors rather than making automated decisions. The six-year graduation rate rose from 32% to 54%; Black and Pell-eligible students now graduate at the same rate as their peers, and GSU produces roughly 2,000 more graduates a year. The difference from the algorithmic-bias cases (35–37) is design: GSU built equity in from the start, used predictions to trigger more human support rather than gatekeeping, and tracked outcomes by demographic group as a primary metric.

THE FULL CASE, IN FIVE BEATS

Background — GSU graduated only a third of students in six years, with large race and income gaps

The Intervention — GSU deployed daily monitoring of 800 risk factors with equity as a primary design constraint

How It Worked — Alerts route through advisors as decision support, delivering proactive outreach rather than gatekeeping

The Evidence — Graduation rose 32 to 54 percent and the equity gap was eliminated rather than narrowed

What Transferred — Construct definition and human-loop architecture, not the model itself, determine whether prediction helps

KEY REFERENCES

- Renick, T. & Strom, A. (2020) on GSU's advising transformation — the system design and outcomes.
- Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data.
- New York Times, "How Colleges Know You're Not Finishing" (2018) — the 800-factor advising model.

LENS APPLICABILITY

LENS treats GSU as the canonical positive case for predictive analytics in education. LEN 4 examines the measurement architecture that made equity a primary outcome. LEN 7 examines the governance structure that kept the system as decision support rather than automated decision. LEN 1 uses it as a problem-framing exemplar.

CASE 42 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

Carnegie Learning's Cognitive Tutor, built from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, is the most rigorously evaluated intelligent tutoring system in education. It uses Bayesian knowledge tracing to model each student's mastery and adapts instruction accordingly, and a RAND Corporation evaluation found statistically significant positive effects on Algebra I achievement. The system is a learning-engineering success in the discipline's own terms — learning science to engineered software to randomized-trial evidence to deployment across 3,000-plus schools. Its limitations are instructive: it works best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the pipeline delivers for problems that fit it — leaving open whether the same discipline can deliver where problems do not.

THE FULL CASE, IN FIVE BEATS

Background — Earlier tutoring systems rested on intuition about learning rather than validated theory or controlled trials

The Intervention — Carnegie Learning built Cognitive Tutor from Anderson's ACT-R architecture with Bayesian knowledge tracing

How It Worked — A decomposable skill model, mastery measurement, and an adaptive interface concentrate practice where weakness sits

The Evidence — RAND's multi-site evaluation found significant Algebra I gains; the program scaled to 3,000-plus schools

What Transferred — Pipeline works for tractable, decomposable problems; ill-structured operational domains remain the open frontier

KEY REFERENCES

- Anderson, J., Corbett, A., Koedinger, K. & Pelletier, R. (1995), "Cognitive Tutors: Lessons Learned," Journal of the Learning Sciences — the ACT-R basis and design.
- Koedinger, K. & Corbett, A. (2006), Cambridge Handbook of the Learning Sciences — knowledge tracing and adaptive instruction.
- RAND Corporation Algebra I evaluation — the statistically significant achievement effects.

LENS APPLICABILITY

LENS uses Cognitive Tutor in LEN 1 as the foundational LE-process exemplar, in LEN 4 as the canonical case for Bayesian knowledge tracing as a measurement instrument, and in LEN 9 as a technical case for model-based adaptive instruction.

CASE 88 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

In 1982, seven people in the Chicago area died after taking Tylenol capsules laced with potassium cyanide. Not knowing who was responsible or how widespread the tampering was, Johnson & Johnson recalled every Tylenol product nationwide — 31 million bottles, at a cost of roughly $100 million — suspended advertising, and engaged openly with the FBI and FDA. The response was unprecedented in U.S. corporate practice, and it was a direct application of the J&J Credo, written in 1943, which had pre-specified that the company's first responsibility was to its customers. The reform that followed reshaped consumer packaging worldwide — tamper-evident seals and blister packs — and the FDA issued tamper-resistant-packaging rules within months. Tylenol recovered its market share within a year.

THE FULL CASE, IN FIVE BEATS

Background — The 1943 Credo pre-specified customers ahead of shareholders, ranking the trade-off crisis pressure inverts

The Intervention — J&J recalled 31 million bottles nationwide and engaged openly with regulators despite localized tampering

How It Worked — Pre-committed values moved the hardest decision out of the moment of maximum pressure

The Evidence — Market share recovered within a year; trust repaid the hundred million spent protecting customers

What Transferred — Tamper-evident packaging became standard and pre-committed values emerged as the deeper institutional principle

KEY REFERENCES

- Kaplan, T. (2014), The Tylenol Crisis — the recall and corporate response.
- James Burke (J&J CEO), in Lasting Leadership (Wharton) — the Credo quote.
- Greyser, S., Johnson & Johnson: The Tylenol Tragedy (HBS case, 1992) — market recovery and crisis management.

LENS APPLICABILITY

LENS uses Tylenol in LEN 7 as the foundational positive case for institutional governance under crisis and in LEN 10 (capstone) as a worked example of pre-committed capability that executed under operational pressure.

CASE 89 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

The Aviation Safety Reporting System, run by NASA on behalf of the FAA since 1976, accepts confidential reports from pilots, controllers, mechanics, and cabin crew about incidents, near-misses, and safety concerns. Its load-bearing feature is institutional, not technical: reporting an event to ASRS confers immunity from FAA enforcement for the conduct reported, within specified limits, making honest reporting the rational choice. Over nearly fifty years and more than two million reports, ASRS has become the world's largest repository of aviation-safety information, surfacing patterns — automation surprise, runway incursions, fatigue effects — before they reached formal investigation thresholds. The architecture has been emulated across domains, and is the canonical success case for an evidence system paired with a credible commitment to non-punitive use.

THE FULL CASE, IN FIVE BEATS

Background — Valuable safety data lives with operators; punitive systems give them strongest reason to stay silent

The Intervention — FAA and NASA created a confidential channel in 1976 conferring immunity for reported conduct

How It Worked — A neutral third party paired with immunity made non-punitive protection credible enough to trust

The Evidence — Over two million reports surfaced automation surprise, runway incursions, and fatigue before accidents

What Transferred — Patient safety, maritime CHIRP, rail, and nuclear systems emulated the protected-reporting design pattern

KEY REFERENCES

- NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume.
- Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased).
- NASA ASRS technical reports (Connell et al.) — patterns first surfaced via ASRS data.

LENS APPLICABILITY

LENS uses ASRS in LEN 4 as the foundational positive case for evidence architecture and in LEN 8 for institutional commitment to non-punitive learning. Studio projects design ASRS-equivalents for new domains.

CASE 90 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

Through whistleblowing and a public inquiry, the Bristol Royal Infirmary was found to have had substantially worse pediatric cardiac-surgery outcomes than other UK centers for years. The Kennedy Inquiry — one of the most influential UK healthcare inquiries in modern times — located the capability gap in the absence of routine specialty-level outcomes reporting: surgeons did not know how their results compared with peers, patients did not know, and referrals did not respond to actual outcome data. The reform built national specialty-level outcomes registries, first in cardiac surgery and then across other specialties, making the UK one of the few countries that routinely publishes surgeon-level results — a paired intervention of technical registry plus institutional commitment to publish that ended outcomes data as the private property of clinicians.

THE FULL CASE, IN FIVE BEATS

Background — No UK system routinely compared pediatric cardiac outcomes; dangerously poor units looked ordinary from inside
The Intervention — The Kennedy Inquiry recommended routine risk-adjusted specialty-level outcomes reporting starting with cardiac surgery
How It Worked — Risk adjustment made cultural acceptance possible by ensuring hard cases would not penalize surgeons
The Evidence — The UK became among few countries publishing surgeon-level results; clinicians themselves became data users
What Transferred — The registry model extended across NHS specialties and pairs with bedside protocols like Keystone

KEY REFERENCES

- Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and recommendations (paraphrased).
- Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports — the registry and surgeon-level publication.
- Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS-safety reform.

LENS APPLICABILITY

LENS uses Bristol in LEN 4 for outcomes-transparency as a paired- intervention deliverable and in LEN 7 for the institutional reform that made surgeon-level publication acceptable. The case pairs with Keystone ICU (Case 14) as healthcare-outcomes interventions at different layers.

CASE 91 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

Singapore Airlines has invested in safety capability across decades in a way that sets it apart from carriers operating under comparable conditions — early adoption of CRM, heavy simulator investment, a young-fleet policy, and a strong reporting culture, sustained even through rapid growth. The 2000 crash of Flight SQ006, which attempted takeoff from a closed, partly-constructed runway at Taipei during a typhoon and killed 83, prompted a sustained institutional self-examination — operational changes, training updates, and public transparency about what had happened — that became a model in the literature on post-accident learning. The airline is the operational successor to Korean Air (Case 23): an Asian carrier that engineered its safety capability deliberately and sustained the investment as a competitive differentiator, not only in response to crisis.

THE FULL CASE, IN FIVE BEATS

Background — Thin aviation margins push capability investment downward; each budget cycle invites trimming the safety margin

The Intervention — From the 1980s Singapore Airlines invested in CRM, simulators, young fleet, and reporting culture

How It Worked — Framing safety as a competitive differentiator tied to brand gave the spend a commercial rationale

The Evidence — SQ006's response chose transparency over defensiveness, becoming a reference case in post-accident learning

What Transferred — Two routes to engineered safety emerge; voluntary investment is harder without crisis as justification

KEY REFERENCES

- Aviation Safety Council (Taiwan), SQ006 Accident Investigation Final Report (2002) — the accident and the airline’s response.
- Singapore Airlines safety reports (multiple years) — training, fleet, and reporting-culture investment.
- IATA Operational Safety Audit (IOSA) documentation — investment ahead of regulatory minimums (paraphrased).

LENS APPLICABILITY

LENS uses Singapore Airlines in LEN 8 for sustained institutional capability investment under competitive pressure. The case pairs with Korean Air (Case 23) as Asian-carrier capability stories of different shapes — one transformation under crisis, the other sustained investment without crisis.

CASE 92 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

Tesla's Autopilot and Full Self-Driving Beta are Level-2 driver-assistance systems: the human driver remains legally and operationally responsible at all times. NHTSA's Standing General Order data has documented dozens of fatal crashes involving Autopilot since the 2016 death of Joshua Brown in Florida, and the pattern is consistent — the system performs capably for long stretches, the driver's monitoring attention attenuates, and an edge case (a stationary object, a faded lane line, a crossing vehicle) produces a collision the inattentive driver fails to catch. NHTSA's investigation has found Tesla's driver-engagement design inadequate to sustain the attention safe operation requires. At consumer scale, Autopilot is the live test of whether passive monitoring of good-enough automation is a role a human can actually perform.

THE FULL CASE, IN FIVE BEATS

The Shift — Level-2 driving automation reaches untrained consumers with no qualification gate for monitoring
What Is Emerging — Fatal crashes accumulate as reliable operation erodes the vigilance the system silently requires
The Capability Question — Liability rests with drivers whose inattention the engagement design itself helped to produce
Early Evidence — NHTSA finds driver-engagement inadequate; automation-complacency evidence points the same direction
Open Problems — Whether any engagement architecture can make consumer Level-2 monitoring sustainable stays open

KEY REFERENCES

- NTSB, Highway Accident Report HAR-17/02 (Williston, FL, 2016) — the quoted disengagement finding.
- NTSB, Highway Accident Report HAR-20/01 (Mountain View, CA, 2018) — Autopilot crash analysis.
- NHTSA Standing General Order 2021-01 reports — documented Autopilot fatal crashes.

LENS APPLICABILITY

LENS uses Tesla Autopilot in LEN 2 as the live consumer-scale test of monitoring as a sustainable role and in LEN 7 for the governance dynamics of a Level-2 system marketed at the boundary of higher autonomy. Studio projects examine the driver-engagement design that would make the role performable.

CASE 93 AUTONOMOUS SYSTEMS 2023

Cruise Robotaxi — Pedestrian Drag

In October 2023, a pedestrian in San Francisco was struck by a human-driven car and thrown into the path of a GM Cruise robotaxi. The robotaxi hit her, detected an impact, and then — executing a pull-to-the-side maneuver — dragged her about twenty feet. The injury was severe, but it was Cruise's institutional response that proved company-ending: regulators found Cruise had initially shown investigators only the first portion of the incident video, omitting the drag. California suspended Cruise's driverless permit, NHTSA opened an investigation, and GM ultimately shut down the commercial operation. The case is the foundational governance-failure case in commercial autonomy: the technology produced one injury; the partial disclosure converted it into the program's collapse. The gap was at governance, not technology.

THE FULL CASE, IN FIVE BEATS

The Shift — Driverless robotaxi oversight rests on operator-controlled disclosure to regulators who cannot inspect directly

What Is Emerging — Pedestrian struck by another car was hit then dragged by a Cruise vehicle

The Capability Question — Whether commercial autonomy programs disclose fully under pressure becomes the institutional question

Early Evidence — Permit suspended; NHTSA opened investigation; GM shut down commercial operations entirely

Open Problems — What auditable pre-committed disclosure commitment commercial autonomy should require before operating

KEY REFERENCES

- California Public Utilities Commission decision suspending Cruise permits (2023) — the omitted-facts finding (paraphrased).
- California DMV order of suspension (2023) — the partial video disclosure.
- Quinn Emanuel report on Cruise incident response (2024) — the disclosure failures.

LENS APPLICABILITY

LENS uses Cruise in LEN 7 as a foundational autonomous-vehicle governance case and in LEN 10 (capstone) for the institutional- response deliverable that should pre-exist any commercial autonomy program.

CASE 94 GOVERNMENT TECHNOLOGY 2016 – PRESENT

COMPAS Recidivism Algorithm

COMPAS is a proprietary risk-assessment algorithm used in some U.S. jurisdictions to inform bail, sentencing, and parole. A 2016 ProPublica investigation found Black defendants were nearly twice as likely as white defendants to be wrongly flagged high-risk (a false-positive rate of about 44.9% versus 23.5%), while white defendants were more often wrongly flagged low-risk. The vendor responded that COMPAS satisfied predictive parity — equal calibration within each group. The literature that followed (Chouldechova 2017; Kleinberg, Mullainathan & Raghavan 2017) proved this an impossibility: when base rates differ, calibration and equal error rates cannot all hold at once. COMPAS surfaced, at criminal-justice scale, that a model can be "accurate" by one metric and inequitable by another — and that choosing the metric is a governance decision, not a technical one.

THE FULL CASE, IN FIVE BEATS

The Shift — Proprietary risk-assessment scores enter bail and sentencing with opacity built into the decision
What Is Emerging — ProPublica found Black defendants twice as likely as whites wrongly flagged high-risk
The Capability Question — Whether fairness has a single technical meaning, or institutions must choose among definitions
Early Evidence — Chouldechova and Kleinberg proved calibration and equal error rates cannot coexist with differing base rates
Open Problems — Distribution of errors is the canonical question; construct definition shapes the trade-off first

KEY REFERENCES

- Angwin, Larson, Mattu & Kirchner, "Machine Bias," ProPublica (2016) — the false-positive-rate disparity (quoted).
- Northpointe technical response (2016) — the predictive-parity (calibration) defense.
- Chouldechova, A. (2017) — the fairness impossibility result for differing base rates.

LENS APPLICABILITY

LENS uses COMPAS in LEN 7 as a foundational algorithmic- accountability case in criminal justice, in LEN 9 for the technical fairness-criterion analysis, and in LEN 4 for the construct-definition question (what counts as recidivism?).

CASE 95 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

FDA-cleared radiology AI tools — for chest X-ray classification, mammography, CT triage — have been repeatedly documented performing worse in deployment than in their validation studies, often with the degradation concentrated in patient groups under-represented in the training data. Larrazabal et al. (PNAS 2020) showed this structurally for chest-X-ray classifiers, and Obermeyer et al. (Science 2019) showed that bias in the labels used to train clinical AI can under-allocate care to Black patients even when the model looks well-calibrated. The FDA's 510(k) clearance process does not routinely require demographic stratification of validation metrics or post-market monitoring of how a tool performs in the population using it. The capability gap is in the regulatory architecture, not the model: clearance is not the same thing as clinically performable deployment.

THE FULL CASE, IN FIVE BEATS

The Shift — Machine-learning diagnostics enter clinical workflow with validation that may not survive deployment

What Is Emerging — Cleared radiology tools repeatedly perform worse in deployment, concentrated in under-represented patient groups

The Capability Question — How a regulator certifies safety without checking behavior across populations and labels it meets

Early Evidence — Degradation reported across imaging, sepsis, dermatology; 510(k) requires no demographic stratification

Open Problems — Capability gap sits in regulatory architecture; demographic post-market surveillance machinery unbuilt

KEY REFERENCES

- FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019) — clearance vs deployment (synthesized).
- Larrazabal et al. (2020), "Gender imbalance in medical imaging datasets," PNAS — sensitivity drops for under-represented groups.
- Obermeyer et al. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," Science — label bias.

LENS APPLICABILITY

LENS uses radiology AI in LEN 4 for clearance-vs-deployment measurement architecture, in LEN 7 for FDA AI/ML regulatory governance, and in LEN 9 for the technical bias-detection pipeline. The case pairs with Vioxx (Case 87) as a post-market surveillance failure pattern at a new technological boundary. The Obermeyer (2019) finding generalizes the diagnosis: bias enters through the labels and through the population, both of which the 510(k) process currently treats as outside its scope.

CASE 96 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

Since ChatGPT's public release in late 2022, the clinical and peer-reviewed literatures have documented a recurring pattern: clinicians use large language models to draft patient education, summaries, or treatment guidance, and the output contains fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe dosing or fabricated contraindications. The capability gap is at the human-verification interface: the model presents hallucinated content with the same fluent confidence as accurate content, and early reports suggest clinicians accept LLM output less critically than a colleague's. The case is the live, foundational case for LLM integration into clinical workflow — the discipline must specify what verification looks like at the moment of use, while deployment is already happening.

THE FULL CASE, IN FIVE BEATS

The Shift — Clinicians adopted ChatGPT immediately at point of care without guidelines or institutional gate

What Is Emerging — Fabricated citations, hallucinated dosages, and fictitious guidelines recur across the documented literature

The Capability Question — Whether clinicians can develop verification routines the model's fluent confidence actively discourages

Early Evidence — Early reports show LLM output accepted less critically than colleague recommendations would be

Open Problems — Routine clinical verification practice analogous to bibliographic discipline remains to be defined

KEY REFERENCES

- JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized).
- Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice" — documented benefits and risks.
- Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024).

LENS APPLICABILITY

LENS uses this case in LEN 2 as the live frontier for human-AI teaming with LLMs, in LEN 7 for the governance of LLM deployment in clinical workflows, and in LEN 10 (capstone) as a prompt for designing verification practices at the moment of use.

CASE 97 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

PredPol and similar predictive-policing tools use historical crime data to forecast where future crime is likely, directing patrols to those locations. Multiple analyses (Lum & Isaac 2016; Richardson, Schultz & Crawford 2019) found that because the historical data records where police have enforced, not where crime has occurred, the algorithm tends to reinforce existing enforcement patterns rather than predict underlying crime — a feedback loop that concentrates policing on already-over-policed neighborhoods. Cities including Santa Cruz, New Orleans, and Los Angeles have since suspended or abandoned predictive-policing deployments after equity reviews. The capability gap is at the construct definition: "where crime occurs" and "where crime is recorded" are different variables, and the tools treated them as one. It is the canonical algorithmic-governance case in U.S. policing.

THE FULL CASE, IN FIVE BEATS

The Shift — Police adopted statistical prediction tools lending discretionary judgment a veneer of objectivity

What Is Emerging — Training on arrest records, models learn enforcement patterns and create self-confirming feedback loops

The Capability Question — Whether a model trained on institutional behavior can predict the underlying phenomenon

Early Evidence — Dirty data documented; Santa Cruz, New Orleans, Los Angeles suspended deployments after equity reviews

Open Problems — Pre-deployment construct-validity audit remains absent in most jurisdictions adopting these systems

KEY REFERENCES

- Lum, K. & Isaac, W. (2016), "To Predict and Serve?," Significance — the enforcement-vs-crime feedback loop (paraphrased).
- Richardson, Schultz & Crawford (2019), "Dirty Data, Bad Predictions" — biased records feeding predictive systems.
- Brantingham et al. (2018) — predictive-policing field experiments.

LENS APPLICABILITY

LENS uses predictive policing in LEN 7 as a foundational AI- governance case in policing and in LEN 9 for the technical construct-validity analysis. The case is paired with COMPAS as criminal-justice algorithmic cases of different kinds.

CASE 98 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the half-century-old protein-structure prediction problem in computational biology. The publicly released AlphaFold Protein Structure Database now contains predicted structures for more than 200 million proteins — essentially the entire known protein universe — and has been integrated into structural-biology and drug-discovery workflows worldwide. AlphaFold is the strongest positive AI case in this book, and its lesson is in the conditions that made the benefit possible: a benchmark (CASP) the field had agreed on for decades, high-quality training data, an output biologists could verify against experimental structures, and an open release that let the global community adopt it fast. Each was a precondition for the success — and none of them was the model itself.

THE FULL CASE, IN FIVE BEATS

The Shift — Deep learning offered computational solution to a fifty-year experimental bottleneck in biology

What Is Emerging — AlphaFold released 200 million predicted structures folded into research workflows worldwide

The Capability Question — Which conditions allow an AI capability to be safely and widely useful

Early Evidence — Success rested on agreed benchmark, clean data, verifiable output, and open release

Open Problems — Whether fields lacking those preconditions can deliberately build them remains the frontier question

KEY REFERENCES

- Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold," Nature — the method and accuracy.
- Varadi et al. (2022), AlphaFold Protein Structure Database, Nucleic Acids Research — the 200M+ structures and open release.
- Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased).

LENS APPLICABILITY

LENS uses AlphaFold in LEN 1 as a problem-framing case for what productive AI deployment looks like, in LEN 9 as a technical achievement, and in LEN 7 for the open-release governance decision that distributed the benefit globally.

CASE 99 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

AI-augmented coding tools — GitHub Copilot, Cursor, Codeium, and peers — represent the largest real-time experiment in human-AI collaboration in this book, with tens of millions of developers using them daily. Published studies (Peng et al. 2023) document real short-term productivity gains; other work (Pearce et al. 2022) finds a substantial share of AI-generated completions in security-sensitive settings contain vulnerabilities, though a controlled study (Sandoval et al. 2023) found AI assistance did not significantly raise the rate of critical security bugs. The capability question is open: are developers becoming more capable, or more dependent? The short-term gains are real; the long-term consequences — especially for those who learn the craft with these tools always available — are not yet known. The discipline is being asked to define good before the longitudinal evidence is in.

THE FULL CASE, IN FIVE BEATS

The Shift — AI coding assistants became infrastructure for tens of millions of developers without controls

What Is Emerging — Productivity gains documented alongside unsettled security findings on AI-generated code quality

The Capability Question — Whether developers using these tools are becoming more capable or more dependent

Early Evidence — Jagged frontier shows performance degrading just outside competence, where users over-trust output

Open Problems — Longitudinal study and training practice that keeps human skill growing remain unbuilt

KEY REFERENCES

- Peng et al. (2023), “The Impact of AI on Developer Productivity” — short-term productivity gains.
- Pearce et al. (2022), “Asleep at the Keyboard? Assessing the Security of GitHub Copilot’s Code Contributions.”
- Sandoval et al. (2023), “Lost at C,” USENIX Security — AI assistance did not significantly increase critical security-bug rates.

LENS APPLICABILITY

LENS uses AI-augmented coding in LEN 2 as the largest live human- AI teaming case in the dataset, in LEN 8 for the long-term capability-development question, and in LEN 10 (capstone) as a prompt for evaluating AI augmentation in professional practice.

CASE 100 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

The ninety-nine cases that precede this one are closed: the investigations are complete, the failures diagnosed, the interventions evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, the graduates who will take positions in healthcare, defense, education at scale, and the human-AI frontier, and the institutions they will build. The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — evidence accumulating, practitioners in training, institutional architecture not yet complete. What the reader does next is what fills in this case.

THE FULL CASE, IN FIVE BEATS

The Shift — The hundredth case belongs to practitioners now in training across the program's domains

What Is Emerging — A teachable discipline of engineerable capability is emerging from training, interfaces, evidence, institutions

The Capability Question — Whether the discipline can be built fast enough to close the documented gaps

Early Evidence — Capability engineering sits where INPO did in 1979 or CRM in 1981

Open Problems — What practitioners build next; a failure prevented, success engineered, institution built

KEY REFERENCES

- LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments.
- Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline’s methods.
- The preceding ninety-nine cases of this volume — the cumulative record of cost and of possibility.

LENS APPLICABILITY

LENS is organized to produce the practitioners who can do this work. The curriculum’s commitments — capability as a system parameter, evidence as a deliverable, ethics as design, implementation as a thread — are the operational form of the argument the ninety-nine preceding cases evidence. Every studio project (LEN 10) is a small attempt at this case.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

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These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization.